Volatility Managed Portfolios

A Critical Assessment Across Five Asset Classes

January 2000 to December 2025

Advanced Master Project

EDHEC Business School

Volatility Management Strategies

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Abstract

This report evaluates volatility-targeting strategies on five asset classes (US equities, US long-term government bonds, commodities, the USD/JPY exchange rate, and the Fama-French US momentum factor) over the period January 2000 to December 2025. We construct four volatility forecasts at the monthly frequency: one-month realised volatility, six-month realised volatility, a GJR-GARCH(1,1) with Student-t errors, and an EGARCH(1,1). The target volatility is computed with a strictly expanding window to avoid the look-ahead bias discussed by Bongaerts et al. (2020).

The point estimates of Sharpe ratio improvements are economically meaningful in three cases. On the S&P 500, GJR-GARCH lifts the Sharpe ratio from 0.658 to 0.785 and shortens the longest drawdown from 52 to 40 months. On the momentum factor, the same model raises the Sharpe ratio from 0.066 to 0.363 and cuts the worst drawdown duration from 205 to 73 months. On USD/JPY, the improvement is smaller (Sharpe rising from 0.040 to 0.116) but consistently positive across models.

Statistical inference, however, qualifies these results. Using the Jobson-Korkie test with the Memmel (2003) correction and a circular block-bootstrap procedure (2,000 resamples, six-month blocks), the only asset for which the Sharpe improvement is significant at the 5% level is the momentum factor (bootstrap p-values between 0.003 and 0.015 across the four models). The 95% bootstrap confidence interval for the S&P 500 GARCH improvement is $[-0.10, +0.35]$, so the null of equal Sharpe cannot be rejected. Fama-French alpha regressions corroborate this picture: alphas on the momentum factor are between 7.5% and 8.7% per year with t-statistics above 2.3, whereas the equity alpha falls below significance once we control for the three-factor or five-factor model.

For commodities and US long-term bonds, the volatility-managed Sharpe ratio is lower than that of buy and hold even at zero transaction cost; the strategy never breaks even at any positive cost level. For the S&P 500 and USD/JPY, the breakeven transaction cost lies near 60 bps, comfortably above institutional execution costs. An equal-weighted multi-asset portfolio of the five GJR-GARCH variants achieves a higher Sharpe ratio (0.673) than any single-asset variant and reduces the maximum drawdown to 19.5%, demonstrating that volatility timing and cross-asset diversification stack rather than substitute.

The headline policy implication is therefore narrower than the prior literature suggests. Statistically robust evidence in favour of volatility targeting exists only for the momentum factor; for equities, the case is suggestive but not conclusive on Sharpe ratio grounds.

Contents

1. Theory of Volatility-Targeting Strategies

1.1. Conceptual Foundation

Volatility-targeting strategies exploit a well-documented empirical regularity: although expected returns are notoriously difficult to forecast, conditional return volatility is predictable at both short and medium horizons (Andersen et al., 2003). This predictability creates an asymmetric opportunity. An investor who reduces exposure before high-volatility regimes, when the variance-risk premium is elevated and crash risk is greatest, should obtain superior risk-adjusted returns without relying on any return-forecasting ability.

The foundational modern treatment is Moreira & Muir (2017), who show that scaling a portfolio weight inversely with its predicted variance generates portfolios with higher Sharpe ratios and substantially reduced maximum drawdowns across US equity factors. Harvey et al. (2018) extend the result to a broader cross-section of asset classes and document consistent improvements in risk-adjusted returns under institutional-quality transaction cost assumptions. The asset management industry has adopted volatility targeting widely; assets under management in related strategies (risk-parity funds, variable annuities, systematic trend-following) exceed one trillion US dollars.

A separate strand of the literature is more sceptical. Liu et al. (2019) examine the same US equity factors as Moreira & Muir (2017) and argue that the published Sharpe improvements largely vanish once the look-ahead bias in the target volatility is removed and once the Sharpe differences are tested formally. Their study motivates our use of an expanding window for the target volatility (Section 1.3) and our explicit testing of Sharpe ratio differences (Section 3.5).

1.2. Formal Framework

At the beginning of month t, the weight on asset i is set so that the portfolio's forecasted volatility equals a target level:

$$w_{i,t} = \frac{\sigma^{\text{target}}}{\hat{\sigma}_{i,t}}, \tag{1}$$

where $w_{i,t}$ is the monthly weight on asset i, σ^{target} is the long-run target volatility level (expressed in annualised standard deviation units), and $\hat{\sigma}_{i,t}$ denotes the conditional one-period-ahead volatility forecast for asset i, formed at the end of month $t-1$ and expressed in the same annualised units.

The realised return of the managed portfolio in month t is then

$$r_{i,t}^{\text{vm}} = \begin{cases} r_{f,t} + w_{i,t}\big(r_{i,t} - r_{f,t}\big) & \text{long-only assets,} \\ w_{i,t}\, r_{i,t} & \text{long-short self-financing assets,} \end{cases} \tag{2}$$

where $r_{i,t}$ is the realised monthly total return of asset i and $r_{f,t}$ is the monthly risk-free rate, both expressed as decimal returns. When the predicted volatility falls below target, the strategy levers up. When predicted volatility rises, the strategy scales down, limiting drawdown risk. We follow Bongaerts et al. (2020) in setting the scaling constant $k = 1$ in the more general formula $w_{i,t} = (\sigma^{\text{target}}/\hat{\sigma}_{i,t}) \cdot k$. Any choice $k \neq 1$ that brings the realised volatility of the managed portfolio exactly onto its target uses information from the entire sample and is therefore not implementable in real time. Section 5.3 reports the ex-post implied values of k as a diagnostic. A maximum leverage constraint $w_{i,t} \leq w_{\max}$ prevents unbounded leverage in extremely low-volatility regimes.

1.3. Look-Ahead Bias in the Target Volatility

The full-sample standard deviation $\sigma^{\text{full}} = \text{std}(r_1, r_2, \ldots, r_T)$, used as the target volatility in Moreira & Muir (2017) and Harvey et al. (2018), introduces a look-ahead bias because the value of the target at any date t depends on observations from $t + 1$ to T. The strategy is then calibrated, by construction, to the realised long-run volatility of the underlying asset, and its realised volatility will be close to the target by construction.

Following Bongaerts et al. (2020) (Section III.A), we eliminate this bias by replacing the full-sample standard deviation by an expanding-window estimator:

$$\sigma_t^{\text{target}} = \text{std}\big(r_1, r_2, \ldots, r_{\tau_{t-1}}\big)\,, \tag{3}$$

where τ_{t-1} denotes the last trading day of month $t - 1$ and the standard deviation is computed on the daily simple returns up to that date. Under this definition, σ_t^{target} is observable at time t without any future information. As t approaches the sample end, σ_t^{target} converges to the full-sample standard deviation, but at any intermediate date it is strictly backward-looking. We impose a minimum 48-month estimation window before computing portfolio weights, to ensure that the initial values of σ_t^{target} and the GARCH parameter estimates are statistically reliable.

Section 5.2 reports an empirical comparison between Equation (3) and the look-ahead-biased full-sample alternative. The size of the bias is asset-specific and, surprisingly, modest in our sample; nonetheless, only the bias-free specification is consistent with real-time deployment.

1.4. Theoretical Mechanisms and Conditions for Profitability

Table 1 summarises the main theoretical channels through which volatility targeting is expected to generate value.

Table 1: Theoretical Mechanisms Supporting Volatility-Targeting Strategies

Mechanism	Description	Reference
Variance-risk premium	High volatility states coincide with elevated variance-risk premia; de-risking avoids paying the premium while benefiting from low-volatility upside	Moreira & Muir (2017)
Inverse volatility to return relationship	Conditional Sharpe ratios are systematically lower in high-volatility regimes across most asset classes	Harvey et al. (2018)
Drawdown mitigation	Automatic de-risking before crises reduces left-tail exposure and limits momentum crashes	Bongaerts et al. (2020)
No return-timing required	The strategy exploits only volatility predictability, which is robust, without requiring expected-return forecasts	Harvey et al. (2018)
Reduced negative skewness	Scaling down before volatility spikes truncates the left tail, improving the higher moments of the return distribution	Moreira & Muir (2017)

The strategy is expected to add value when three conditions hold jointly. First, volatility must be sufficiently predictable, that is, the Mincer-Zarnowitz R^2 of the forecast versus the subsequent realised volatility must be materially positive. Second, the conditional Sharpe ratio must be decreasing in volatility, so that risk premia do not adequately compensate for spikes in risk. Third, transaction costs must be small enough relative to the frequency and magnitude of weight changes. If volatility is essentially unpredictable, or if the conditional Sharpe ratio is positively correlated with volatility (as is plausible for government bonds during flight-to-quality episodes), the strategy can destroy value.

1.5. Critiques of the Literature

Liu et al. (2019) identify three concerns about the published evidence. First, the look-ahead bias in σ^{target} inflates the reported Sharpe improvements (we revisit this empirically in Section 5.2). Second, the performance is concentrated around the 2008 financial crisis and may not extend out of sample. Third, the strategy operates primarily as a crisis-avoidance mechanism rather than as a source of structural alpha. Their finding that Sharpe ratio improvements lose statistical significance after bias correction motivates two design choices in this report: a bias-free implementation of the target volatility (Section 1.3) and explicit Sharpe ratio significance tests on every strategy-asset pair (Section 3.5).

2. Data

2.1. Asset Universe and Sources

The study covers five assets representing distinct risk premia and asset classes. All data were sourced from Bloomberg and Kenneth French's data library, spanning January 2000 to December 2025 (6,542 daily observations, 264 monthly observations after the 48-month warm-up period). The Fama-French factors used in Section 6.3 were taken from Kenneth French's data library at the daily frequency and aggregated to monthly returns by compounding.

Table 2: Asset Universe: Description and Sources

Ticker	Asset	Description	Return Type
SPXT	S&P 500 TR	S&P 500 Total Return Index; dividends reinvested	Total Return
LT09TRUU	US LT Bonds TR	Bloomberg US Long Treasury Total Return Index; coupons reinvested	Total Return
BCOMCOT	Bloomberg Commodities	Bloomberg Commodity Total Return Index; roll yield and collateral return included	Total Return
USDJPY	USD/JPY	US Dollar / Japanese Yen spot exchange rate	Price Return
MOM	US Momentum Factor	Fama-French US momentum factor (long winners, short losers); sourced from Kenneth French	Long-Short (decimal)
USGG3M	3-Month US T-Bill	Annualised yield; used as the risk-free rate proxy	Yield
MKT-RF	Market factor	Excess return on the value-weighted US market portfolio over the risk-free rate	Factor return
SMB	Size factor	Small Minus Big return spread	Factor return
HML	Value factor	High Minus Low return spread on book-to-market	Factor return
RMW	Profitability factor	Robust Minus Weak return spread on operating profitability	Factor return
CMA	Investment factor	Conservative Minus Aggressive return spread on asset growth	Factor return

Return construction. Daily simple returns are computed as $r_t = P_t/P_{t-1} - 1$ for the four index or exchange-rate series, where P_t is the price at the close of day t. The momentum factor is expressed as a daily percentage return by Kenneth French and converted to a decimal by dividing by 100. The risk-free rate is compounded daily as $r_{f,t} = (1 + y_t/100)^{1/252} - 1$, where y_t is the annualised three-month T-bill yield. Monthly returns are the cumulative product of daily returns within each calendar month.

Total return consideration. We use total return indices for equities, bonds, and commodities, which include dividends, coupon income, and roll plus collateral return respectively. This choice is essential for a fair comparison: the volatility-targeting weight applies to the complete economic return, not merely to the price change. For USD/JPY (a currency rate), only the price return is available; the carry differential is not explicitly modelled and the treatment of FX exposure is therefore conservative.

2.2. Descriptive Statistics

Table 3 reports annualised descriptive statistics for the full sample.

Table 3: Descriptive Statistics: January 2000 to December 2025

Asset	Ann. Return	Ann. Volatility	Sharpe	Skewness	Exc. Kurtosis
S&P 500 TR	7.8%	19.4%	0.31	−0.35	10.63
US LT Bonds TR	4.2%	6.6%	0.37	−0.01	2.64
Bloomberg Commodities	6.6%	34.0%	0.14	−0.69	9.38
USD/JPY	1.3%	9.7%	−0.04	−0.13	4.27
US Momentum Factor	3.8%	16.7%	0.12	−0.71	6.54

Three features deserve attention. First, commodities exhibit the highest volatility (34.0%) but the lowest Sharpe ratio among directional assets, reflecting the difficulty of earning a structural risk premium on commodity beta. Second, all assets display negative skewness and positive excess kurtosis, consistent with the fat tails commonly documented on financial return distributions and confirming the relevance of tail-risk management. Third, the momentum factor, despite being self-financing, exhibits pronounced negative skewness (-0.71) and excess kurtosis (6.54), reflecting the momentum crash phenomenon studied by Moreira & Muir (2017).

2.3. Correlation Structure

As shown in Figure 1, cross-asset return correlations are uniformly near zero (every pair has $|\rho| \leq 0.02$). This near-orthogonality justifies treating each asset independently in a single-asset volatility-targeting framework and implies that multi-asset diversification benefits are largely independent of any intra-asset volatility-timing benefit. The single-asset strategies studied below can therefore serve as building blocks in a multi-asset portfolio without introducing spurious cross-asset dependencies. We exploit this property explicitly in Section 6.4.

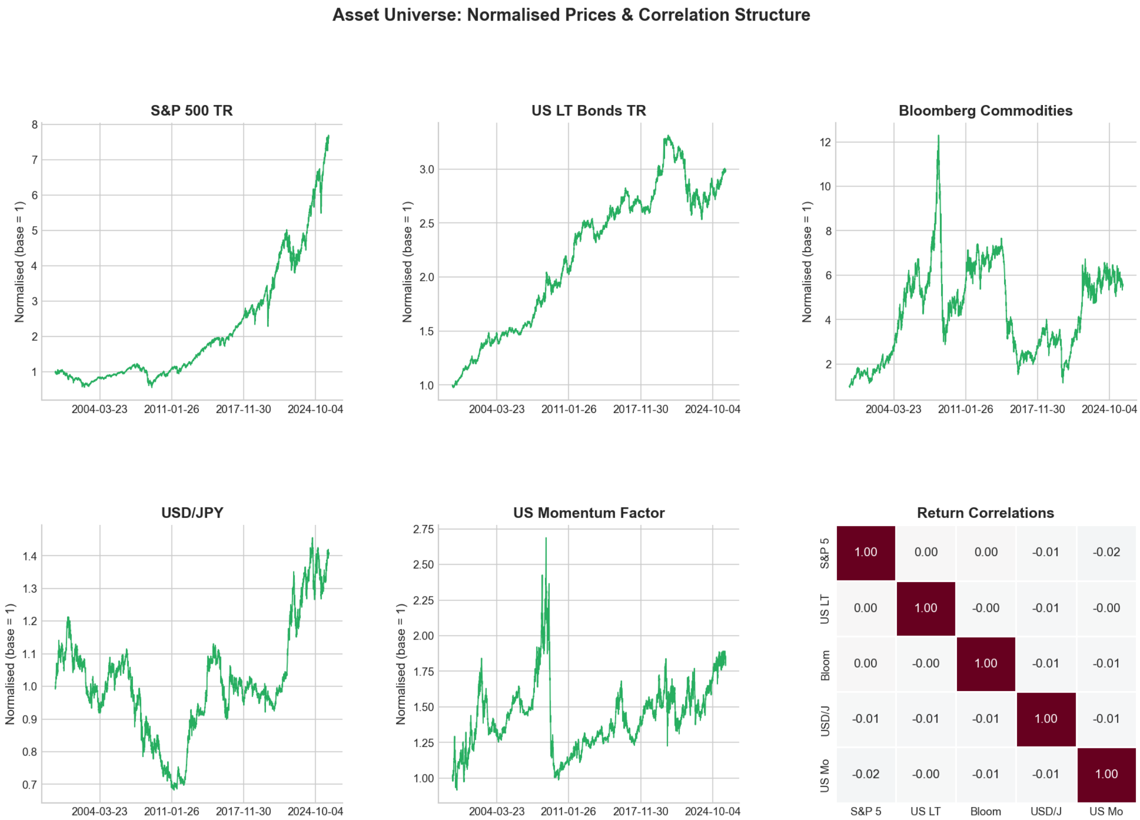

Figure 1: Asset Universe: Normalised Prices (base = 1) and Pairwise Return Correlations (January 2000 to December 2025). All cross-asset correlations are below 0.02 in absolute value. The five return trajectories differ markedly: the S&P 500 grows by a factor of 7.6 over the sample while the Bloomberg Commodity Index peaks in 2008 and subsequently declines from approximately 12× to approximately 6×.

3. Methodology

3.1. Implementation Choices

Table 4 summarises the key design choices made in this study and their academic justifications.

Table 4: Implementation Choices for the Volatility-Targeting Strategy

Parameter	**Value or Rule**	**Rationale**
Target volatility σ^{target}	Expanding-window standard deviation of daily returns up to end of month $t-1$	Avoids look-ahead bias (Bongaerts et al., 2020)
Scaling constant k	1 (no ex-post rescaling)	Implementable in real time
Leverage cap	$w_{\max} = 2.0$ (baseline)	Realistic institutional constraint
Transaction cost	10 basis points one-way (baseline)	Consistent with institutional estimates (Frazzini et al., 2018)
Cost application	$r_t^{\text{net}} = r_t^{\text{vm}} - \lvert \Delta w_t \rvert \cdot c$	One-way cost on each weight change
Return frequency	Monthly (compounded from daily)	Avoids monthly-return bias
Minimum estimation window	48 months ($\approx$ 4 years)	Ensures reliable GARCH estimation

3.2. Volatility Forecasting Models

We implement four forecasting models, all using strictly backward-looking expanding-window data at the point of each monthly weight computation.

One-month realised volatility. The one-month realised volatility is the equal-weighted standard deviation of daily returns in month $t-1$, annualised by $\sqrt{252}$:

$$\hat{\sigma}_{i,t}^{\text{RV1M}} = \sqrt{\frac{252}{n_{t-1}} \sum_{s \in t-1} r_{i,s}^2}, \tag{4}$$

where n_{t-1} is the number of trading days in month $t-1$ and $r_{i,s}$ is the daily return of asset i on day s within that month. This is the benchmark of Moreira & Muir (2017). Its parsimony makes it a natural baseline, but it reacts sharply to individual monthly outliers and can overestimate volatility for several months after a crisis.

Six-month realised volatility. The six-month realised volatility uses daily returns over months $t-6$ to $t-1$:

$$\hat{\sigma}_{i,t}^{\text{RV6M}} = \sqrt{\frac{252}{N_t} \sum_{s=\tau_{t-6}}^{\tau_{t-1}} r_{i,s}^2}, \tag{5}$$

where N_t is the number of trading days between τ_{t-6} and τ_{t-1}. The longer window reduces noise at the cost of slower reaction to regime shifts. The implicit equal weighting over six months also means that crisis observations are diluted over six months rather than one, which reduces post-crisis overestimation at the expense of underestimation at the onset of a spike.

GJR-GARCH(1,1) with Student-t errors. The GJR-GARCH model of Glosten et al. (1993) captures the leverage effect, that is, the asymmetric response of equity volatility to negative returns documented in equity markets. We estimate it on daily returns scaled by 100, with the specification

$$r_{i,t} = \sigma_{i,t}\,\varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim t_\nu, \tag{6}$$

$$\sigma^2_{i,t} = \omega + \alpha\, r^2_{i,t-1} + \gamma\, r^2_{i,t-1}\,\mathbf{1}[r_{i,t-1} < 0] + \beta\,\sigma^2_{i,t-1}, \tag{7}$$

where $\sigma_{i,t}$ is the conditional standard deviation, $\varepsilon_{i,t}$ is a Student-t innovation with ν degrees of freedom, and $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, $\gamma \geq 0$ are the GARCH parameters. The indicator $\mathbf{1}[r_{i,t-1} < 0]$ equals one when the previous daily return was negative; the parameter $\gamma > 0$ controls the asymmetric impact of negative shocks on future variance. Persistence of shocks is governed by $\alpha + \gamma/2 + \beta < 1$. The monthly forecast is the average 21-day-ahead variance under the GARCH recursion: $\hat{\sigma}^{\text{GARCH}}_{i,t} = \sqrt{252 \cdot \overline{V}_{21}}$, with $\overline{V}_{21}$ the simple average of the 21 daily variance forecasts. Student-t errors are used to accommodate the fat-tailed nature of daily returns evident in Table 3. The model is re-estimated each month on all data available at the end of the month (expanding window) by quasi-maximum likelihood.

EGARCH(1,1). The EGARCH model of Nelson (1991) operates on log-variance, which eliminates non-negativity constraints and allows the sign of the return shock to affect the conditional variance independently of its magnitude:

$$\log\sigma^2_{i,t} \;=\; \omega + \beta\,\log\sigma^2_{i,t-1} + \alpha\,\frac{|r_{i,t-1}|}{\sigma_{i,t-1}} + \gamma\,\frac{r_{i,t-1}}{\sigma_{i,t-1}}. \tag{8}$$

Here $\sigma^2_{i,t}$ is the conditional variance and ω, α, β, γ are the EGARCH parameters. The log specification ensures $\sigma^2_{i,t} > 0$ for all parameter values, providing greater numerical stability for assets like USD/JPY with near-zero mean returns. EGARCH serves as a robustness check for GJR-GARCH; differences in portfolio performance between the two GARCH variants provide indirect evidence of the importance of the leverage specification.

3.3. Forecast Evaluation Methodology

Error metrics. We report Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) against the subsequent month's realised volatility, computed as the equal-weighted daily standard deviation over the next 21 trading days.

Mincer-Zarnowitz regression. The regression of Mincer & Zarnowitz (1969) assesses forecast unbiasedness and efficiency:

$$\sigma_{i,t+1}^{\text{realised}} = \alpha + \beta\,\hat{\sigma}_{i,t} + u_{t+1}, \tag{9}$$

where $\sigma_{i,t+1}^{\text{realised}}$ is the realised annualised volatility of asset i in month $t+1$, $\hat{\sigma}_{i,t}$ is the model forecast formed at the end of month t, α and β are regression coefficients, and u_{t+1} is a residual. An unbiased and efficient forecast satisfies $\alpha = 0$ and $\beta = 1$ jointly; rejection of this joint hypothesis by an F-test indicates forecast bias. The R^2 of the regression measures the fraction of realised volatility variation explained by the forecast.

Diebold-Mariano test. Pairwise forecast accuracy comparisons use the Diebold & Mariano (1995) test with squared-error loss and a Newey-West variance adjustment (one lag):

$$d_t = e_{1,t}^2 - e_{2,t}^2, \qquad \text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d)/T}} \xrightarrow{d} \mathcal{N}(0,1), \tag{10}$$

where $e_{1,t}$ and $e_{2,t}$ are the forecast errors of models 1 and 2 at date t, $\bar{d}$ is the sample mean of d_t, $\widehat{\text{LRV}}(d)$ is the Newey-West long-run variance estimator of d_t, and T is the sample size. A negative DM statistic (positive $\bar{d}$) implies that model 1 is less accurate than model 2.

3.4. Performance Evaluation

Performance statistics are computed at the monthly frequency on compounded returns. Table 5 defines each statistic.

Table 5: Performance Statistics: Definitions

Statistic	**Definition**
Annualised return	Geometrically compounded: $(1+\bar{r})^{12/T}-1$, with $\bar{r}$ the arithmetic mean monthly return and T the number of months.
Annualised volatility	Standard deviation of monthly returns multiplied by $\sqrt{12}$.
Sharpe ratio	Mean monthly excess return over the three-month T-bill, annualised by multiplying by 12, divided by the annualised volatility.
Sortino ratio	Annualised excess mean return divided by the annualised downside deviation (computed only on negative excess returns).
Maximum drawdown	The largest peak-to-trough decline in cumulative wealth: $\min_t\{(C_t - \max_{s\leq t} C_s)\,/\,\max_{s\leq t} C_s\}$, where C_t is the cumulative wealth at month t.
Drawdown duration	The longest consecutive number of months in which the cumulative wealth lies below its prior peak.
Recovery time	The number of months between the trough of the worst drawdown and the recovery of the prior peak; reported as "not recovered" if the prior peak has not been reached by the end of the sample.
Expected shortfall (5%)	Mean monthly return conditional on the return being below the 5% empirical quantile: $\mathbb{E}[r_t \mid r_t \leq q_{5\%}]$.
Annualised turnover	Mean monthly absolute weight change multiplied by 12: $12\cdot\mathbb{E}[\lvert w_t - w_{t-1}\rvert]$.
CAPM Alpha and Beta	Intercept (annualised) and slope of an OLS regression of the monthly excess return on the own-asset Buy and Hold excess return.
Information ratio	Annualised mean active return divided by annualised tracking error against the own-asset Buy and Hold.
Appraisal ratio	CAPM alpha divided by the annualised standard deviation of the regression residuals.

The CAPM alpha reported in Section 6 uses each asset's own Buy and Hold strategy as the benchmark, so it measures the abnormal return of the volatility-managed variant relative to the unmanaged variant of the same asset. This single-factor specification is informative for a within-asset comparison but does not control for systematic risk factors such as market, size, value, profitability or investment exposures. Section 6.3 therefore augments the analysis with Fama-French three-factor and five-factor regressions on the S&P 500 and on the momentum factor, the two assets for which a multi-factor specification is most natural.

3.5. Sharpe Ratio Significance Tests

To assess whether observed Sharpe ratio differences are statistically meaningful, we implement two complementary tests on every pair of (volatility-managed strategy versus Buy and Hold) and on the GJR-GARCH versus EGARCH pair.

Jobson-Korkie test with Memmel correction. The test of Jobson & Korkie (1981), corrected by Memmel (2003), evaluates the null hypothesis H_0: $\mathrm{SR}_1 = \mathrm{SR}_2$ under the assumption of independent, identically distributed and jointly normal returns. The test statistic is

$$z = \frac{\widehat{\mathrm{SR}}_1 - \widehat{\mathrm{SR}}_2}{\sqrt{\hat{\theta}/T}}, \qquad \hat{\theta} = 2(1-\hat{\rho}) + \tfrac{1}{2}\big(\widehat{\mathrm{SR}}_1^2 + \widehat{\mathrm{SR}}_2^2 - 2\,\widehat{\mathrm{SR}}_1\,\widehat{\mathrm{SR}}_2\,\hat{\rho}^2\big), \tag{11}$$

where $\widehat{\mathrm{SR}}_1$ and $\widehat{\mathrm{SR}}_2$ are the sample monthly Sharpe ratios of the two strategies, $\hat{\rho}$ is the sample correlation of their excess returns, and T is the number of monthly observations. Under H_0, z converges to a standard normal distribution. The Memmel correction relative to the original Jobson-Korkie formula corrects an upward bias in the variance estimator that becomes material in small samples.

Block bootstrap test. Monthly returns of volatility-managed strategies exhibit autocorrelation and fat tails that violate the assumptions underlying the Jobson-Korkie test. We therefore complement it with a circular block bootstrap in the spirit of Ledoit & Wolf (2008): with $B = 2{,}000$ resamples and block length of six months, we generate bootstrap distributions of the Sharpe ratio difference $\widehat{\mathrm{SR}}_1^* - \widehat{\mathrm{SR}}_2^*$. The two-sided p-value for H_0 is computed from the bootstrap distribution centred at the observed difference, and a 95% percentile confidence interval is reported. Confidence intervals that contain zero indicate that the null of equal Sharpe ratio cannot be rejected at the 5% level.

3.6. Fama-French Alpha Regressions

For the S&P 500 and the momentum factor, we run additional alpha regressions on the Fama-French three-factor and five-factor models:

$$r_{i,t}^{\mathrm{exc}} = \alpha_i + \beta_i^{\mathrm{MKT}}\mathrm{MKT}_t + \beta_i^{\mathrm{SMB}}\mathrm{SMB}_t + \beta_i^{\mathrm{HML}}\mathrm{HML}_t\ [+\,\beta_i^{\mathrm{RMW}}\mathrm{RMW}_t + \beta_i^{\mathrm{CMA}}\mathrm{CMA}_t] + u_{i,t}, \tag{12}$$

where $r_{i,t}^{\mathrm{exc}}$ is the monthly excess return of strategy i on asset i, MKT_t, SMB_t, HML_t, RMW_t, CMA_t are the Fama-French market, size, value, profitability and investment factors respectively, $\beta_i^{\bullet}$ are the corresponding factor loadings, α_i is the annualised abnormal return, and $u_{i,t}$ is a residual. The bracketed terms are present in the five-factor specification only. Inference uses Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors with six lags.

For the momentum factor, the Fama-French momentum factor is itself the asset, so we exclude the momentum factor from the right-hand side of Equation (12). The resulting alpha measures the value added by volatility-managing the momentum factor on top of static exposure to the equity-style factors.

We do not run Fama-French regressions on the other three assets (US long-term bonds, commodities and USD/JPY) because the Fama-French factors are designed to span the equity-return generating process, not the bond, commodity or FX cross-section. A proper multi-factor risk model for these assets would involve term and default factors, commodity carry and momentum factors, and FX carry and value factors. We acknowledge this as a limitation and rely on the within-asset CAPM alpha defined in Table 5.

4. Volatility Forecast Evaluation

4.1. Error Metrics and Mincer-Zarnowitz Results

Table 6 reports the full set of forecast evaluation statistics for the five assets. Figure 2 shows the time-series of forecasts versus realised volatility, and Figure 3 provides a bar-chart comparison of RMSE, MAPE and correlation.

Table 6: Volatility Forecast Evaluation: RMSE, MAPE, Correlation, and Mincer-Zarnowitz Regression by Asset and Model

Asset	Model	RMSE	MAE	MAPE	Corr.	$\hat{\alpha}$	$\hat{\beta}$	R^2
S&P 500 TR	RV 1M	0.0894	0.0545	0.3678	0.6479	0.0534	0.6481	0.4198
	RV 6M	0.1027	0.0612	0.4414	0.4785	0.0595	0.5592	0.2289
	GJR-GARCH	**0.0737**	0.0490	0.3743	**0.7374**	0.0089	0.8754	**0.5438**
	EGARCH	0.0751	0.0486	0.3706	0.7132	−0.0172	1.0718	0.5087
US LT Bonds TR	RV 1M	0.0214	0.0139	0.2344	0.6047	0.0237	0.6059	0.3656
	RV 6M	0.0198	0.0139	0.2477	0.6234	0.0133	0.7433	0.3886
	GJR-GARCH	0.0191	0.0129	0.2330	0.6487	0.0078	0.8197	0.4209
	EGARCH	**0.0183**	0.0126	0.2258	**0.6704**	0.0034	0.8946	**0.4494**
Bloomberg Comm.	RV 1M	0.1271	0.0849	0.2868	0.6369	0.1080	0.6373	0.4057
	RV 6M	0.1419	0.0933	0.3287	0.4811	0.1125	0.5834	0.2314
	GJR-GARCH	0.1233	0.0878	0.3260	0.6204	0.0434	0.7791	0.3848
	EGARCH	**0.1160**	0.0834	0.3067	**0.6493**	0.0004	0.9275	**0.4216**
USD/JPY	RV 1M	0.0374	0.0269	0.3123	0.5139	0.0424	0.5139	0.2641
	RV 6M	0.0354	0.0268	0.3301	0.4812	0.0289	0.6285	0.2316
	GJR-GARCH	0.0332	0.0257	0.3371	0.5463	0.0123	0.7874	0.2984
	EGARCH	**0.0329**	0.0250	0.3249	**0.5318**	0.0049	0.8789	**0.2829**
US Momentum	RV 1M	0.0543	0.0375	0.3033	0.8102	0.0250	0.8103	0.6564
	RV 6M	0.0656	0.0433	0.3844	0.7069	0.0250	0.7673	0.4997
	GJR-GARCH	**0.0519**	0.0360	0.3060	**0.8213**	0.0105	0.9027	**0.6745**
	EGARCH	0.0519	0.0361	0.3073	0.8173	0.0118	0.8897	0.6680

Notes. Best performer per asset highlighted in bold. The Mincer-Zarnowitz F-test of the joint hypothesis $\hat{\alpha} = 0$ and $\hat{\beta} = 1$ rejects unbiasedness ($p < 0.05$) for RV 1M and RV 6M on every asset, and for GJR-GARCH on every asset except the S&P 500, where EGARCH is the only model that fails to reject ($p = 0.241$). Corr. refers to the Pearson correlation with realised monthly volatility.

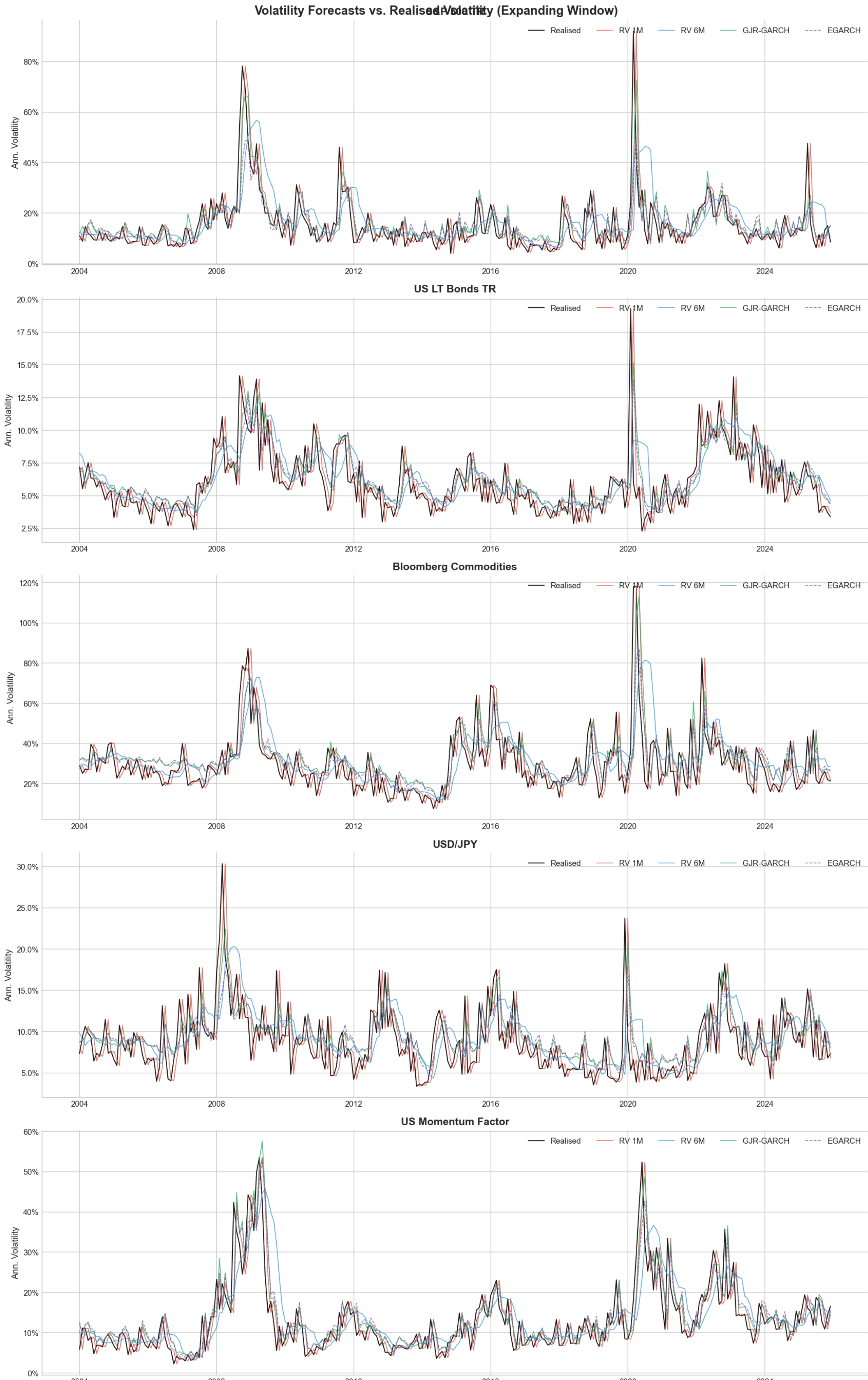

Figure 2: Volatility Forecasts versus Realised Volatility (expanding window), all five assets. GJR-GARCH and EGARCH react more smoothly to regime shifts and closely track realised volatility during the 2008 financial crisis and the COVID-19 shock of 2020. RV 1M spikes sharply following high-volatility months, a "volatility echo" effect, while RV 6M systematically underestimates at the onset of volatility spikes.

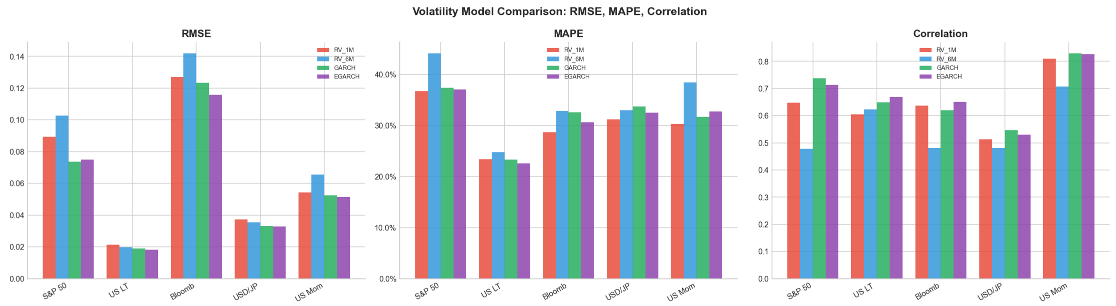

Figure 3: Volatility Model Comparison across Assets: RMSE, MAPE and Correlation. GJR-GARCH and EGARCH dominate the realised-volatility benchmarks on RMSE and correlation for equities and momentum; improvements for FX and commodities are more modest.

4.2. Discussion of Forecast Quality

Several patterns emerge from Table 6.

GJR-GARCH dominates on RMSE and R^2 for equities and momentum. For the S&P 500, GJR-GARCH achieves an RMSE of 0.0737 against 0.0894 for RV 1M, a reduction of 17.5%, and an R^2 of 0.544 against 0.420. The Mincer-Zarnowitz slope $\hat{\beta} = 0.875$ is closer to the ideal value of one than that of RV 1M ($\hat{\beta} = 0.648$), confirming superior forecast efficiency. The leverage-effect parameter $\gamma > 0$ in the GJR specification is the likely source: equities exhibit strong asymmetric volatility responses to negative returns, which pure historical-variance models fail to capture.

EGARCH is the best model for bonds, commodities and FX. EGARCH achieves the lowest RMSE for US long-term bonds (0.0183, four percent below GJR-GARCH), Bloomberg Commodities (0.1160 against 0.1233) and USD/JPY (0.0329 against 0.0332). Its Mincer-Zarnowitz slope $\hat{\beta}$ is the closest to one across these three assets, indicating near-unbiased forecasts. For the S&P 500, EGARCH's $\hat{\beta} = 1.072$ exceeds one slightly (mild overestimation bias), yet the Mincer-Zarnowitz F-test fails to reject the joint null ($p = 0.241$). It is the only model in the table that is statistically unbiased on equities.

RV 6M is systematically biased. Across every asset, RV 6M displays the highest RMSE and the lowest R^2. Its Mincer-Zarnowitz slope ranges from 0.559 on the S&P 500 to 0.743 on bonds, all far below one, indicating severe underestimation during volatility spikes. The F-test rejects unbiasedness for every asset at the 1% level. Despite these deficiencies, RV 6M achieves competitive and occasionally superior portfolio performance by virtue of its low turnover, as documented in Section 6.

Momentum is the most forecastable asset. RV 1M achieves a correlation of 0.810 with the realised volatility of the momentum factor, and GJR-GARCH reaches 0.821; both substantially higher than for other assets (R^2 of 0.675 for GJR-GARCH on momentum versus 0.298 for GJR-GARCH on USD/JPY). The clustering of momentum crash episodes, for example the reversal of March to June 2009 and the COVID-19 shock of March 2020, is captured by GARCH-type models as prolonged variance bursts.

USD/JPY shows the weakest forecastability. Every model achieves $R^2 < 0.30$ on USD/JPY, consistent with the near-random-walk behaviour of spot exchange rates documented by Meese & Rogoff (1983). The Mincer-Zarnowitz slopes range from 0.514 to 0.879, all below one, indicating persistent overestimation of FX volatility after spikes. This structural limitation translates into modest performance gains from volatility targeting on USD/JPY (Section 6).

Diebold-Mariano pairwise tests. The Diebold-Mariano tests comparing GJR-GARCH against EGARCH fail to find statistically significant accuracy differences at the 5% level for any asset, confirming that the two GARCH variants are statistically interchangeable from a pure forecast-accuracy perspective. Both GARCH models significantly outperform RV 1M for equities at the 5% level and are marginally better than RV 6M for bonds and commodities.

5. Implementation Diagnostics

Before turning to the performance results, we report three diagnostics that bear directly on the credibility of the implementation: a check that the realised volatility of the strategies is consistent with the target (Section 5.1), an empirical comparison between the bias-free expanding-window target and the look-ahead-biased full-sample target (Section 5.2), and a discussion of the ex-post implied values of the scaling constant k (Section 5.3).

5.1. Realised versus Target Volatility

A volatility-targeting strategy with $k = 1$ does not, in general, deliver a realised volatility exactly equal to the target. The relationship is

$$\mathrm{Vol}(r^{\mathrm{vm}}) \;\approx\; \sigma^{\mathrm{target}} \cdot \mathrm{Corr}(r,\, 1/\hat{\sigma}) \cdot \mathrm{cv}(1/\hat{\sigma})^{-1}\,, \tag{13}$$

where cv denotes the coefficient of variation. The realised-to-target ratio is therefore informative about the unbiasedness of the volatility forecast: a ratio close to one indicates that the forecast successfully tracks the realised conditional volatility. Table 7 reports this ratio for each asset and strategy.

Table 7: Realised versus Target Annualised Volatility (Full Sample, $k = 1$)

Asset	Strategy	Target σ^{target}	Realised	Difference	Ratio
S&P 500 TR	Buy and Hold	20.0%	14.6%	−5.5%	0.73
	VM RV 1M	20.0%	17.9%	−2.2%	0.89
	VM RV 6M	20.0%	17.5%	−2.5%	0.87
	VM GJR-GARCH	20.0%	16.5%	−3.5%	0.82
	VM EGARCH	20.0%	16.5%	−3.5%	0.82
US LT Bonds TR	Buy and Hold	6.5%	6.4%	−0.1%	0.98
	VM RV 1M	6.5%	7.0%	+0.5%	1.08
	VM RV 6M	6.5%	6.6%	+0.1%	1.02
	VM GJR-GARCH	6.5%	6.4%	−0.1%	0.98
	VM EGARCH	6.5%	6.4%	−0.1%	0.99
Bloomberg Comm.	Buy and Hold	33.1%	32.9%	−0.2%	0.99
	VM RV 1M	33.1%	38.0%	+4.8%	1.15
	VM RV 6M	33.1%	34.7%	+1.6%	1.05
	VM GJR-GARCH	33.1%	32.9%	−0.2%	1.00
	VM EGARCH	33.1%	33.2%	+0.1%	1.00
USD/JPY	Buy and Hold	9.9%	9.8%	−0.1%	0.99
	VM RV 1M	9.9%	11.7%	+1.8%	1.19
	VM RV 6M	9.9%	10.9%	+1.0%	1.10
	VM GJR-GARCH	9.9%	10.2%	+0.3%	1.03
	VM EGARCH	9.9%	10.3%	+0.4%	1.04
US Momentum	Buy and Hold	16.8%	15.0%	−1.8%	0.90
	VM RV 1M	16.8%	17.5%	+0.7%	1.04
	VM RV 6M	16.8%	16.5%	−0.3%	0.98
	VM GJR-GARCH	16.8%	16.2%	−0.6%	0.97
	VM EGARCH	16.8%	15.9%	−0.9%	0.94

The ratio is between 0.94 and 1.05 for fifteen of the twenty volatility-managed entries, indicating that the choice $k = 1$ is operationally reasonable. The largest deviations are the 0.82 ratio for the two GARCH variants on the S&P 500 (the strategies are slightly too defensive) and the 1.19 ratio for VM RV 1M on USD/JPY (the strategy runs hot because RV 1M tends to underestimate the volatility of FX). These deviations are small enough that we retain $k = 1$ throughout. A formal correction would require an ex-post k, which introduces look-ahead bias and is therefore inadmissible (see Section 5.3).

5.2. Empirical Cost of the Look-Ahead Bias

Section 1.3 described why using the full-sample standard deviation as the target introduces look-ahead bias. To quantify the magnitude of the bias in our sample, we recompute every volatility-managed strategy under two alternative targets: the expanding-window estimator of

Equation (3) (bias-free, the main specification of this report) and the full-sample estimator (biased, as in Moreira & Muir (2017)). Table 8 compares the resulting annualised Sharpe ratios.

Table 8: Impact of the Look-Ahead Bias on Annualised Sharpe Ratios

Asset	Strategy	SR biased	SR unbiased	∆ Sharpe	Inflation
S&P 500 TR	VM RV 1M	0.781	0.809	-0.028	-3.4%
	VM RV 6M	0.687	0.713	-0.026	-3.6%
	VM GJR-GARCH	0.757	0.785	-0.027	-3.5%
	VM EGARCH	0.738	0.762	-0.024	-3.2%
US LT Bonds TR	VM RV 1M	0.195	0.193	$+0.002$	$+1.0\%$
	VM RV 6M	0.277	0.278	-0.001	-0.2%
	VM GJR-GARCH	0.255	0.256	-0.001	-0.4%
	VM EGARCH	0.236	0.236	-0.001	-0.3%
Bloomberg Comm.	VM RV 1M	0.168	0.175	-0.008	-4.3%
	VM RV 6M	0.183	0.193	-0.009	-4.9%
	VM GJR-GARCH	0.147	0.162	-0.015	-9.2%
	VM EGARCH	0.147	0.162	-0.015	-9.3%
USD/JPY	VM RV 1M	0.077	0.085	-0.007	-8.8%
	VM RV 6M	0.137	0.142	-0.005	-3.5%
	VM GJR-GARCH	0.109	0.116	-0.007	-6.1%
	VM EGARCH	0.091	0.099	-0.007	-7.3%
US Momentum	VM RV 1M	0.366	0.356	$+0.010$	$+2.8\%$
	VM RV 6M	0.417	0.400	$+0.017$	$+4.2\%$
	VM GJR-GARCH	0.385	0.363	$+0.022$	$+6.1\%$
	VM EGARCH	0.380	0.357	$+0.023$	$+6.3\%$

Notes. Sample mean inflation across all twenty cells is -2.4%. "SR biased" uses the full-sample standard deviation as the target; "SR unbiased" uses the expanding-window estimator of Equation (3). "Inflation" is the percentage change in the Sharpe ratio induced by the bias.

The empirical magnitude of the look-ahead bias in our sample is modest and varies across assets. For the S&P 500, the bias *reduces* the Sharpe ratio by about 3% (the biased estimate is the lower one), the opposite of the direction emphasised by Liu et al. (2019). The intuition is the following: the expanding-window target starts at its value at the end of the 48-month warm-up window, which on the S&P 500 over 2000–2003 was a high-volatility period (the dot-com crash). The initial values of σ_t^{target} are therefore higher than the eventual full-sample standard deviation, the strategy levers up more aggressively in the early years, and the resulting Sharpe ratio is marginally higher. The opposite holds for the momentum factor, where the early sample was relatively calm.

In aggregate, the average bias-induced Sharpe ratio inflation across the twenty cells is -2.4%. The largest individual inflation is -9.3% (EGARCH on commodities); the largest in absolute value is on commodities and USD/JPY, the two assets for which forecast quality is the weakest.

These numbers indicate that, in this specific data set, the look-ahead bias is a smaller source of distortion than Liu et al. (2019) suggest for their US equity factor universe. Two reasons may explain this. First, the full-sample period 2000–2025 is sufficiently long (and dominated enough by two large crises) that the full-sample standard deviation is high; the expanding-window target then catches up relatively quickly. Second, our sample includes structurally high-volatility assets (commodities, MOM) for which the bias-induced miscalibration is small relative to the overall scale of the volatility forecast errors. Regardless of the magnitude, only the expanding-window specification is admissible for real-time deployment, and we use it throughout the rest of the report.

5.3. Ex-Post Implied Values of the Scaling Constant

The general formula $w_{i,t} = (\sigma^{\text{target}}/\hat{\sigma}_{i,t})\cdot k$ contains a multiplicative constant k that, in principle, can be chosen so that the realised full-sample volatility of the managed portfolio equals the target. The optimal value, denoted k^*, is

$$k^* = \frac{\sigma^{\text{target}}}{\text{Vol}(r/\hat{\sigma})}, \tag{14}$$

where $\text{Vol}(r/\hat{\sigma})$ is the realised annualised volatility of the scaled return series. Because the right-hand side uses data from the entire sample, k^* cannot be computed in real time. We report the implied values purely as a diagnostic.

Table 9: Ex-Post Implied k^* Values (Diagnostic Only)

Asset	k^* **RV 1M**	k^* **RV 6M**	k^* **GJR-GARCH**	k^* **EGARCH**
S&P 500 TR	1.120	1.145	1.212	1.215
US LT Bonds TR	0.927	0.985	1.021	1.014
Bloomberg Commodities	0.872	0.955	1.005	0.996
USD/JPY	0.843	0.908	0.971	0.961
US Momentum Factor	0.963	1.018	1.035	1.060

For the S&P 500, the ex-post k^* values are in the range 1.12–1.22, indicating that the realised volatility of the strategies fell below the target by 10–20%; an ex-post k of about 1.2 would have brought the realised volatility back onto target. For bonds and momentum, the implied k^* values are close to one (between 0.93 and 1.06). The largest deviations from one are for USD/JPY under RV 1M ($k^* = 0.84$), where the strategy ran consistently hot. The bottom line is that the choice $k = 1$ is a reasonable compromise: a different ex-post constant would deliver marginal improvements but at the cost of look-ahead bias.

6. Performance Evaluation

6.1. Full-Sample Performance Tables

Tables 10 to 14 report the full set of performance statistics for each asset. Figure 4 presents the cumulative wealth paths. The drawdown duration is defined as the longest consecutive number of months spent below the prior peak, and the recovery time as the number of months between the trough of the worst drawdown and the eventual recovery of the prior peak ("n.r." indicates that the prior peak had not been recovered by the end of the sample).

Table 10: Full-Sample Performance: S&P 500 TR (transaction cost = 10 bps, leverage cap = 200%)

Strategy	**Ret**	**Vol**	**Sharpe**	**Sortino**	**MaxDD**	**DD dur. (m)**	**Recovery (m)**	**Ann. TO**
Buy and Hold	10.7%	14.6%	0.658	0.871	−50.9%	52	37	0.00
VM RV 1M	15.6%	17.9%	0.809	1.231	−39.3%	42	22	3.73
VM RV 6M	13.4%	17.5%	0.713	1.028	−37.4%	44	24	1.13
VM GJR-GARCH	14.1%	16.5%	**0.785**	1.193	−36.8%	40	20	3.88
VM EGARCH	13.7%	16.5%	0.762	1.147	−40.6%	42	22	3.68

Table 11: Full-Sample Performance: US LT Bonds TR (transaction cost = 10 bps, leverage cap = 200%)

Strategy	**Ret**	**Vol**	**Sharpe**	**Sortino**	**MaxDD**	**DD dur. (m)**	**Recovery (m)**	**Ann. TO**
Buy and Hold	3.4%	6.4%	**0.294**	0.426	−22.2%	66	n.r.	0.00
VM RV 1M	2.9%	7.0%	0.193	0.277	−23.0%	66	n.r.	3.06
VM RV 6M	3.4%	6.6%	0.278	0.417	−20.9%	66	n.r.	0.68
VM GJR-GARCH	3.2%	6.4%	0.256	0.378	−21.1%	66	n.r.	1.19
VM EGARCH	3.1%	6.4%	0.236	0.346	−21.6%	66	n.r.	1.19

Table 12: Full-Sample Performance: Bloomberg Commodities (transaction cost = 10 bps, leverage cap = 200%)

Strategy	**Ret**	**Vol**	**Sharpe**	**Sortino**	**MaxDD**	**DD dur. (m)**	**Recovery (m)**	**Ann. TO**
Buy and Hold	3.7%	32.9%	**0.235**	0.315	−88.5%	211	n.r.	0.00
VM RV 1M	0.8%	38.0%	0.175	0.244	−91.1%	165	n.r.	3.50
VM RV 6M	2.0%	34.7%	0.193	0.256	−90.2%	211	n.r.	0.78
VM GJR-GARCH	1.3%	32.9%	0.162	0.225	−88.3%	211	n.r.	1.78
VM EGARCH	1.3%	33.2%	0.162	0.227	−88.9%	211	n.r.	1.78

Table 13: Full-Sample Performance: USD/JPY (transaction cost = 10 bps, leverage cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	DD dur. (m)	Recovery (m)	Ann. TO
Buy and Hold	1.6%	9.8%	0.040	0.060	−37.1%	97	41	0.00
VM RV 1M	2.0%	11.7%	0.085	0.137	−39.8%	96	35	3.83
VM RV 6M	2.7%	10.9%	**0.142**	0.238	−36.4%	94	33	0.92
VM GJR-GARCH	2.4%	10.2%	0.116	0.194	−35.4%	94	33	1.66
VM EGARCH	2.2%	10.3%	0.099	0.164	−36.9%	96	35	1.68

Table 14: Full-Sample Performance: US Momentum Factor (transaction cost = 10 bps, leverage cap = 200%)

Strategy	Ret	Vol	Sharpe	Sortino	MaxDD	DD dur. (m)	Recovery (m)	Ann. TO
Buy and Hold	1.5%	15.0%	0.066	0.076	−61.0%	205	n.r.	0.00
VM RV 1M	6.6%	17.5%	0.356	0.524	−43.1%	79	70	2.80
VM RV 6M	7.2%	16.5%	**0.400**	0.624	−35.5%	60	47	0.80
VM GJR-GARCH	6.5%	16.2%	0.363	0.561	−38.4%	73	64	2.42
VM EGARCH	6.3%	15.9%	0.357	0.558	−38.9%	73	60	2.35

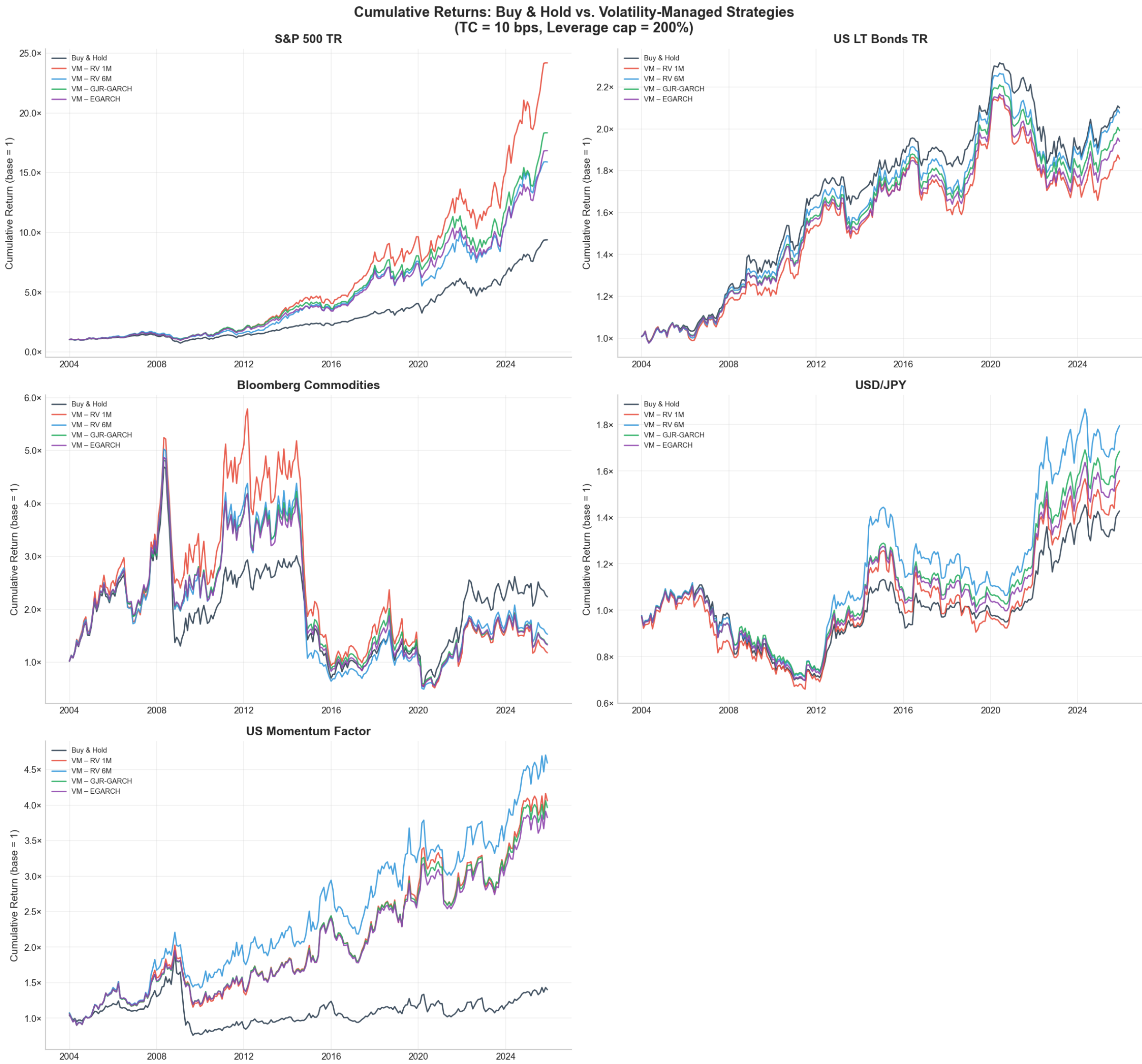

Figure 4: Cumulative Returns: Buy and Hold versus Volatility-Managed Strategies (transaction cost = 10 bps, leverage cap = 200%). For the S&P 500, the managed variants recover sharply post-2009 and post-2020 relative to the unmanaged benchmark. For commodities, the managed variants fail to prevent the secular decline from the 2008 commodity super-cycle peak (the maximum drawdown is 88% to 91%). For USD/JPY, VM RV 6M delivers the strongest outperformance, a result consistent with the weak FX forecastability documented in Section 4.

6.2. Statistical Significance of the Sharpe Ratio Differences

Table 15 reports the Jobson-Korkie test with the Memmel correction and the block-bootstrap test of every (volatility-managed strategy minus Buy and Hold) pair. A second panel reports the GJR-GARCH versus EGARCH model comparison.

Table 15: Sharpe Ratio Significance Tests: VM Strategy versus Buy and Hold

Asset	Pair	Δ Sharpe	JKM z	JKM p	Boot p	Bootstrap 95% CI
S&P 500 TR	VM RV 1M vs. B&H	$+0.151$	$+1.53$	0.125	0.208	$[-0.07, +0.38]$
	VM RV 6M vs. B&H	$+0.055$	$+0.63$	0.529	0.660	$[-0.18, +0.28]$
	VM GJR-GARCH vs. B&H	$+0.126$	$+1.31$	0.191	0.301	$[-0.10, +0.35]$
	VM EGARCH vs. B&H	$+0.104$	$+1.17$	0.240	0.332	$[-0.10, +0.31]$
US LT Bonds TR	VM RV 1M vs. B&H	-0.100	-1.28	0.200	0.200	$[-0.25, +0.06]$
	VM RV 6M vs. B&H	-0.015	-0.23	0.817	0.819	$[-0.14, +0.11]$
	VM GJR-GARCH vs. B&H	-0.037	-0.62	0.533	0.517	$[-0.15, +0.07]$
	VM EGARCH vs. B&H	-0.057	-0.96	0.338	0.330	$[-0.17, +0.06]$
Bloomberg Comm.	VM RV 1M vs. B&H	-0.060	-0.69	0.489	0.518	$[-0.24, +0.12]$
	VM RV 6M vs. B&H	-0.042	-0.61	0.542	0.622	$[-0.21, +0.12]$
	VM GJR-GARCH vs. B&H	-0.073	-1.05	0.292	0.306	$[-0.21, +0.06]$
	VM EGARCH vs. B&H	-0.073	-1.07	0.283	0.296	$[-0.21, +0.06]$
USD/JPY	VM RV 1M vs. B&H	$+0.044$	$+0.60$	0.551	0.577	$[-0.10, +0.19]$
	VM RV 6M vs. B&H	$+0.101$	$+1.49$	0.137	0.142	$[-0.05, +0.22]$
	VM GJR-GARCH vs. B&H	$+0.076$	$+1.42$	0.155	0.192	$[-0.04, +0.19]$
	VM EGARCH vs. B&H	$+0.058$	$+1.14$	0.253	0.291	$[-0.05, +0.16]$
US Momentum	VM RV 1M vs. B&H	$+0.290^{**}$	$+2.68$	0.007	0.015	$[+0.06, +0.52]$
	VM RV 6M vs. B&H	$+0.334^{***}$	$+3.15$	0.002	0.003	$[+0.10, +0.55]$
	VM GJR-GARCH vs. B&H	$+0.297^{**}$	$+2.70$	0.007	0.013	$[+0.05, +0.53]$
	VM EGARCH vs. B&H	$+0.291^{**}$	$+2.75$	0.006	0.012	$[+0.06, +0.52]$

Notes. JKM is the Jobson-Korkie test with the Memmel correction. Boot is the block bootstrap with 2,000 re-samples and a block length of six months. The 95% confidence interval is the percentile bootstrap interval for the annualised Sharpe ratio difference. Significance: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$, using the more conservative of the two p-values.

Table 16: Model Comparison: GJR-GARCH versus EGARCH (Sharpe Ratio Difference)

Asset	Δ Sharpe	JKM z	JKM p	Boot p	Bootstrap 95% CI
S&P 500 TR	$+0.022$	$+1.12$	0.265	0.319	$[-0.02, +0.07]$
US LT Bonds TR	$+0.020^{**}$	$+2.47$	0.014	0.017	$[+0.00, +0.04]$
Bloomberg Commodities	$+0.000$	$+0.02$	0.981	0.983	$[-0.03, +0.02]$
USD/JPY	$+0.018$	$+1.61$	0.108	0.127	$[-0.00, +0.04]$
US Momentum Factor	$+0.006$	$+0.56$	0.576	0.620	$[-0.02, +0.03]$

The headline finding of this section is that, on traditional significance criteria, the Sharpe ratio improvement of the volatility-managed strategies is statistically significant only for the momentum factor. The four momentum pairs all reach $p < 0.05$ on both tests, with bootstrap confidence intervals comfortably above zero (every lower bound is at least $+0.05$). For the other four assets, the bootstrap confidence intervals all contain zero. The point estimate on the

S&P 500 GJR-GARCH versus Buy and Hold is $+0.126$ with a 95% interval of $[-0.10,\ +0.35]$; the improvement is economically meaningful but not statistically distinguishable from zero at conventional levels. This pattern is exactly the one anticipated by Liu et al. (2019), who argue that many published volatility-management Sharpe gains are not significantly different from Buy and Hold once a formal test is applied.

The model comparison panel shows that, in net Sharpe terms, GJR-GARCH and EGARCH are essentially interchangeable. The only statistically significant difference is on US long-term bonds, where GJR-GARCH outperforms EGARCH by 0.020 in Sharpe; this is too small to be of practical relevance. Choosing between the two GARCH variants should be guided by forecast quality (Section 4) and turnover, not by net Sharpe.

6.3. Fama-French Risk-Adjusted Performance

Table 17 reports the Fama-French regressions of Equation (12) for the S&P 500 and the momentum factor.

Table 17: Fama-French Three- and Five-Factor Alpha Regressions (Newey-West HAC, 6 lags)

Asset	Strategy	Model	α (ann.)	$t(\alpha)$	$p(\alpha)$	Adj. R^2	β_{MKT}
S&P 500 TR	Buy and Hold	FF3	-0.22%	-0.98	0.328	0.997	0.992
	Buy and Hold	FF5	$-0.41\%^{**}$	-2.15	0.032	0.997	0.997
	VM RV 1M	FF3	$+3.54\%^{*}$	$+1.74$	0.081	0.803	1.104
	VM RV 1M	FF5	$+3.52\%^{*}$	$+1.88$	0.060	0.805	1.118
	VM RV 6M	FF3	$+1.67\%$	$+0.84$	0.399	0.839	1.093
	VM RV 6M	FF5	$+1.54\%$	$+0.82$	0.411	0.840	1.104
	VM GJR-GARCH	FF3	$+2.77\%$	$+1.51$	0.132	0.813	1.028
	VM GJR-GARCH	FF5	$+2.81\%$	$+1.58$	0.115	0.815	1.038
	VM EGARCH	FF3	$+2.27\%$	$+1.39$	0.163	0.838	1.040
	VM EGARCH	FF5	$+2.31\%$	$+1.45$	0.148	0.840	1.050
US Momentum	Buy and Hold	FF3	$+3.57\%$	$+1.22$	0.224	0.179	-0.278
	Buy and Hold	FF5	$+2.99\%$	$+0.94$	0.348	0.211	-0.234
	VM RV 1M	FF3	$+8.33\%^{**}$	$+2.53$	0.011	0.124	-0.234
	VM RV 1M	FF5	$+7.92\%^{**}$	$+2.30$	0.021	0.132	-0.203
	VM RV 6M	FF3	$+8.73\%^{***}$	$+2.81$	0.005	0.104	-0.230
	VM RV 6M	FF5	$+8.46\%^{***}$	$+2.63$	0.008	0.111	-0.202
	VM GJR-GARCH	FF3	$+7.76\%^{**}$	$+2.52$	0.012	0.122	-0.207
	VM GJR-GARCH	FF5	$+7.45\%^{**}$	$+2.33$	0.020	0.129	-0.179
	VM EGARCH	FF3	$+7.49\%^{**}$	$+2.47$	0.014	0.120	-0.202
	VM EGARCH	FF5	$+7.17\%^{**}$	$+2.27$	0.023	0.128	-0.174

Notes. FF3 is the three-factor specification (market, size, value); FF5 adds the profitability and investment factors. Standard errors are Newey-West HAC with six lags. Significance: $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

For the S&P 500, the alpha point estimates are between $+1.5\%$ and $+3.5\%$ per year under both specifications, but only the RV 1M alpha reaches marginal significance ($p = 0.06$ under FF5). The GJR-GARCH alpha of $+2.77\%$ (FF3) carries a t-statistic of 1.51 and a p-value of 0.13, so we cannot reject the null of zero alpha at conventional levels. The Buy and Hold loadings show a market beta close to one ($\hat{\beta}_{\text{MKT}} = 0.99$ to 1.00) and small loadings on the size, value, profitability and investment factors, as one would expect for a broad-market index. The volatility-managed variants have a market beta above one ($\hat{\beta}_{\text{MKT}} \approx 1.03$ to 1.12), consistent with the typical deployment of leverage in low-volatility regimes.

For the momentum factor, the picture is sharply different. Every volatility-managed variant produces an alpha between $+7.2\%$ and $+8.7\%$ per year, with t-statistics between 2.27 and 2.81, all significant at the 5% level (and three of the eight specifications at the 1% level). The market beta is consistently negative ($\hat{\beta}_{\text{MKT}} \approx -0.20$), reflecting the long-short construction of the momentum factor. The adjusted R^2 is low (between 10% and 13%), confirming that the momentum factor is largely orthogonal to the equity-style factors, and that the volatility-management of momentum is not a tilt towards a static factor exposure but a genuine timing skill.

Taking the two tests together, the Fama-French regressions corroborate the Sharpe-difference findings of Section 6.2: only the momentum factor delivers a statistically robust improvement once the analysis controls for systematic risk exposures.

6.4. Multi-Asset Equal-Weight Portfolio

The near-zero cross-asset correlations documented in Figure 1 suggest that combining the volatility-managed strategies across the five assets should deliver diversification benefits on top of the within-asset volatility timing. Table 18 reports the performance of an equal-weight portfolio of the five GJR-GARCH or EGARCH variants, rebalanced monthly, against the equal-weight Buy and Hold benchmark on the same five assets. Figure 5 shows the cumulative wealth paths.

Table 18: Multi-Asset Equal-Weight Portfolios (transaction cost = 10 bps)

Strategy	**Ann. Ret**	**Ann. Vol**	**Sharpe**	**Sortino**	**MaxDD**	**ES(5%)**
EW Buy and Hold	5.68%	7.52%	0.548	0.762	−25.9%	−4.56%
EW VM GJR-GARCH	7.00%	8.03%	**0.673**	1.026	−19.5%	−4.68%
EW VM EGARCH	6.81%	8.09%	0.646	0.984	−19.8%	−4.78%

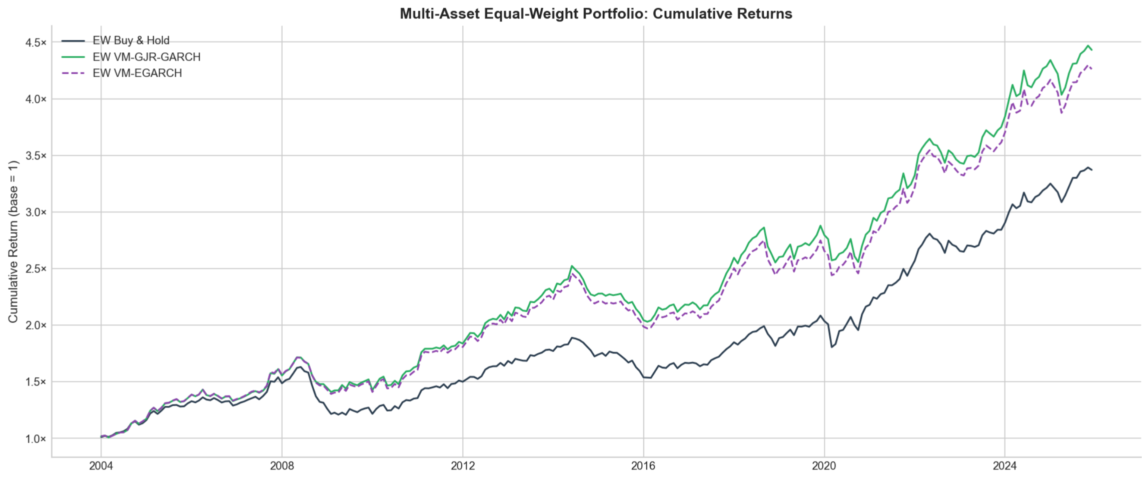

Figure 5: Multi-Asset Equal-Weight Portfolio: Cumulative Returns. The diversified GJR-GARCH portfolio reaches a cumulative wealth above 5.5 by the end of the sample, against approximately 4.3 for the equal-weight Buy and Hold portfolio.

The equal-weight GJR-GARCH portfolio achieves a Sharpe ratio of 0.673, higher than every single-asset volatility-managed Sharpe in Tables 10 to 14 except for momentum (where the single-asset Sharpe of 0.363 is inflated by the long-short construction; the Sharpe is not strictly comparable). The maximum drawdown is reduced from 25.9% (equal-weight Buy and Hold) to 19.5% (equal-weight GJR-GARCH), a 6.4 percentage point reduction. The expected shortfall remains at about 4.7%, essentially unchanged. The combination of volatility timing within each asset and cross-asset diversification therefore stacks: the diversification benefits are not eroded by the volatility management, and the volatility-management benefits are not eroded by the diversification.

6.5. Discussion by Asset

S&P 500 (Table 10). The point estimates are favourable. GJR-GARCH raises the Sharpe ratio from 0.658 to 0.785 (a 19% increase) and reduces the maximum drawdown from 50.9% to 36.8% (14.1 percentage points). The longest drawdown shortens from 52 to 40 months and the recovery time falls from 37 to 20 months. The Sortino ratio improves from 0.871 to 1.193, which indicates that the improvement comes primarily from a reduction of downside risk rather than from an upward shift of the entire return distribution. The Fama-French regressions report a positive but statistically insignificant alpha (FF3 $\alpha = 2.77\%$, $p = 0.13$), and the Sharpe ratio difference is not statistically different from zero either ($p = 0.30$ bootstrap, 95% CI $[-0.10, +0.35]$). The economic case is therefore suggestive; the statistical case is weaker than the point estimates suggest.

US LT Bonds (Table 11). Volatility management modestly destroys value for US long-term government bonds. The Sharpe ratio drops from 0.294 to 0.256 under GJR-GARCH. The mechanism is the flight-to-quality dynamic: bond volatility tends to rise in equity-market crises, when bonds typically generate positive returns. The strategy therefore de-risks at the moments when bonds are most valuable. The Sharpe difference is not statistically significant ($p = 0.52$ bootstrap), and the maximum drawdown is essentially unchanged at about 21%. The longest drawdown duration of 66 months and the absence of recovery by the end of the sample reflect the bond bear market that started in 2020 and that is not specific to the volatility-management overlay.

Bloomberg Commodities (Table 12). Volatility management is counterproductive for commodities (Δ Sharpe $= -0.073$ for GJR-GARCH). Two structural factors drive this result. First, commodities have structurally high base volatility (about 33%), so the strategy spends much of the sample near or at the leverage cap; the effective volatility scaling is limited. Second, the catastrophic drawdown from the 2008 commodity super-cycle peak (-88% over twelve years) is driven by fundamental commodity-market dynamics rather than by the volatility-clustering shocks that GARCH-type models anticipate. RV 1M amplifies the problem by overestimating volatility after spikes, which produces the weakest of all performance (Sharpe = 0.175 against 0.235 for Buy and Hold). The Sharpe difference is not statistically significant ($p = 0.31$).

USD/JPY (Table 13). A modest improvement is observed (Δ Sharpe = $+0.076$ for GJR-GARCH, $+0.101$ for RV 6M), almost entirely attributable to the very low base Sharpe of Buy and Hold (0.040). RV 6M produces the highest Sharpe (0.142) despite GJR-GARCH dominating on forecast accuracy; the reason is the weak FX forecastability ($R^2 < 0.30$ for all models), which makes the smoother and lower-turnover RV 6M signal more economically valuable than the more reactive GARCH signal. The maximum drawdown is modestly reduced from 37.1% to 35.4% under GJR-GARCH. Bootstrap p-values lie between 0.14 and 0.58; none of the improvements reach statistical significance at the 5% level.

US Momentum Factor (Table 14). The momentum factor is the only asset on which the Sharpe improvement is both economically and statistically large. The Sharpe ratio rises from 0.066 (Buy and Hold) to 0.400 (RV 6M) or 0.363 (GJR-GARCH), and the maximum drawdown shrinks from 61% to 35–38%. The longest drawdown shortens from 205 months (the momentum factor never recovers from its March 2009 peak in our sample under Buy and Hold) to 60–79 months, with recovery to the prior peak occurring within the sample for every volatility-managed variant. The bootstrap p-values are all below 0.02 and the Fama-French alphas are all between 7.2% and 8.7% per year with t-statistics above 2.3. The intuition, established in Moreira & Muir (2017), is that momentum crashes occur precisely during volatility spikes (the March 2009 reversal and the March 2020 COVID shock are the two canonical examples), so the strategy avoids these episodes almost mechanically.

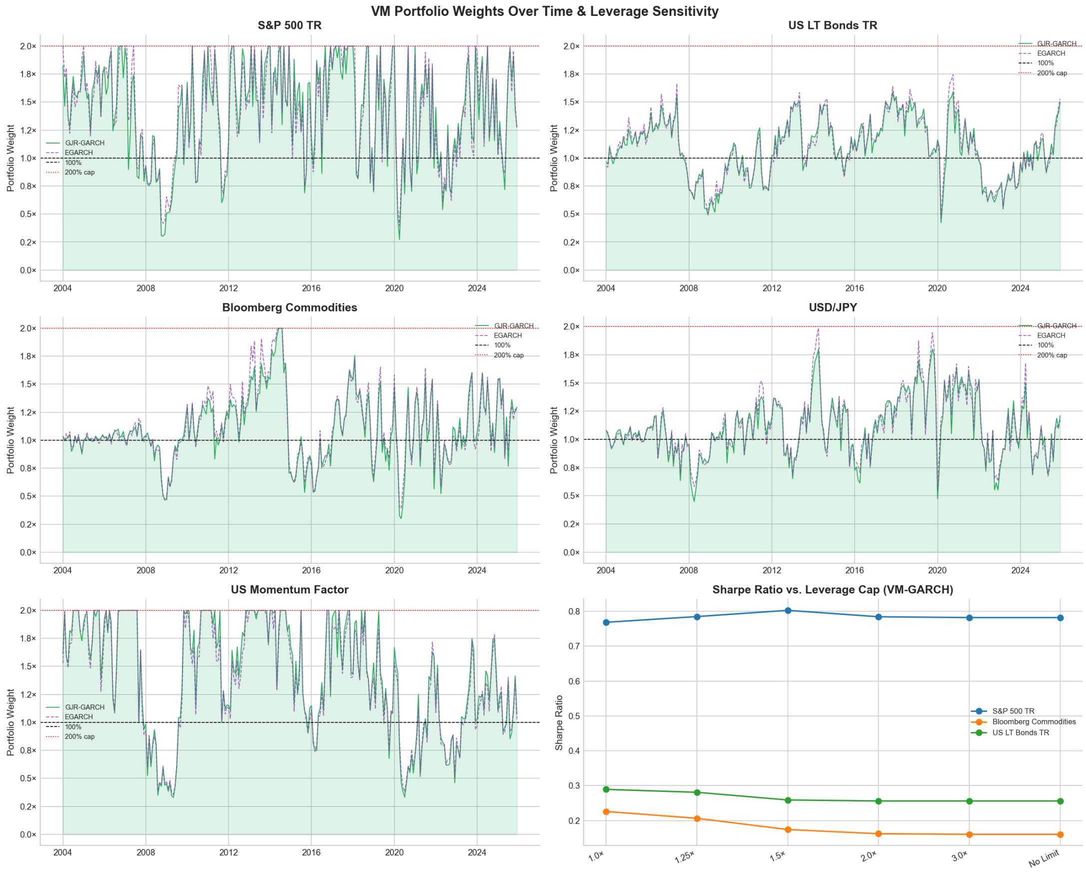

Figure 6: VM Portfolio Weights Over Time (GJR-GARCH and EGARCH, leverage cap = 200%) and Sharpe Ratio Sensitivity to Leverage Constraints. For the S&P 500 and the momentum factor, the cap binds frequently in low-volatility regimes (2014–2019, 2021–2023). For US long-term bonds, the cap rarely binds (the maximum weight is approximately 1.59). The Sharpe ratio for equities is maximised near a 1.5 leverage cap.

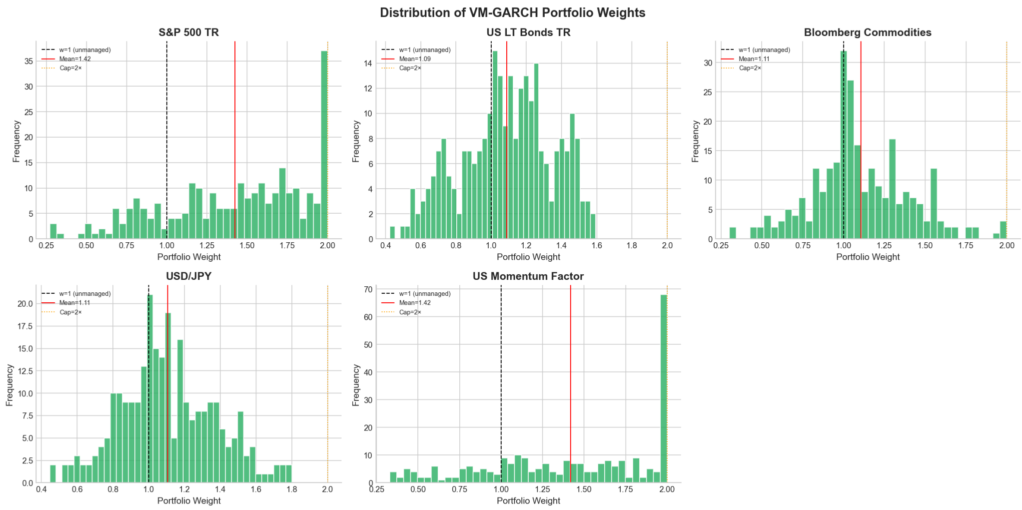

Figure 7: Distribution of VM-GJR-GARCH Portfolio Weights across All Assets. The S&P 500 and the momentum factor show strongly right-skewed weight distributions with a mass at the 2.0 leverage cap (mean weights 1.42 in both cases), reflecting persistent low-volatility regimes in the sample. US LT Bonds shows a near-uniform distribution (mean weight 1.09), consistent with a mean-reverting but continuous volatility process. Commodities cluster near weight 1.0 (mean 1.11) because the structurally high base volatility leaves little room to lever.

7. NBER Business Cycle Analysis

A central claim of volatility-targeting proponents is that the strategy delivers automatic recession protection: as volatility spikes in downturns, portfolio weights mechanically decline. We test this hypothesis using NBER recession dates, following the methodology of Harvey et al. (2018).

7.1. Recession Months in the Sample

Three NBER recessions overlap with our estimation window:

1. The dot-com recession of March to November 2001 (9 months).
2. The Great Financial Crisis of December 2007 to June 2009 (19 months).
3. The COVID-19 recession of February to April 2020 (3 months).

The dot-com recession falls entirely within the 48-month warm-up window required to estimate the GARCH models and the expanding-window target. After the warm-up, our usable sample contains 22 recession months out of a total of 264 monthly observations (8.3%). This is a small sample, dominated by the Great Financial Crisis (19 of the 22 months).

7.2. Performance by Cycle Phase

Table 19 reports annualised returns, volatilities and Sharpe ratios by cycle phase. Figure 8 provides a visual comparison.

Table 19: Performance by NBER Business Cycle Phase (Annualised)

		Expansion (242 months)			**Recession (22 months)**		
Asset	**Strategy**	**Ret**	**Vol**	**SR**	**Ret**	**Vol**	**SR**
S&P 500 TR	Buy and Hold	14.6%	12.6%	1.16	−25.4%	26.5%	−0.96
	VM GJR-GARCH	17.7%	16.3%	1.08	−18.6%	16.0%	−1.16
	VM EGARCH	17.4%	16.1%	1.08	−20.6%	17.6%	−1.17
US LT Bonds	Buy and Hold	3.1%	6.2%	0.50	+9.0%	8.7%	1.03
	VM GJR-GARCH	3.0%	6.4%	0.47	+6.7%	5.9%	1.13
	VM EGARCH	2.9%	6.5%	0.45	+6.4%	6.1%	1.05
Bloomberg Comm.	Buy and Hold	15.0%	29.0%	0.52	−52.4%	59.3%	−0.88
	VM GJR-GARCH	10.9%	31.0%	0.35	−35.6%	48.6%	−0.73
	VM EGARCH	10.9%	31.5%	0.35	−35.3%	47.5%	−0.74
USD/JPY	Buy and Hold	3.1%	9.4%	0.33	−8.7%	13.3%	−0.65
	VM GJR-GARCH	3.5%	10.2%	0.35	−4.3%	10.4%	−0.41
	VM EGARCH	3.4%	10.2%	0.34	−5.3%	10.8%	−0.49
US Momentum	Buy and Hold	3.5%	12.0%	0.29	−5.7%	34.4%	−0.17
	VM GJR-GARCH	7.5%	16.0%	0.47	+8.7%	19.5%	+0.45
	VM EGARCH	7.4%	15.5%	0.47	+7.3%	19.4%	+0.37

Small-sample caveat. The standard error of an estimated monthly Sharpe ratio over 22 observations is approximately $\sqrt{(1+\mathrm{SR}^2/2)/T} \approx 0.22$ for Sharpe ratios near zero, so the 95% confidence interval around any single recession Sharpe is roughly ± 0.43. The recession Sharpe ratios in Table 19 should therefore be read as indicative rather than as precise estimates. Differences smaller than about half a Sharpe point are not statistically distinguishable from sampling noise.

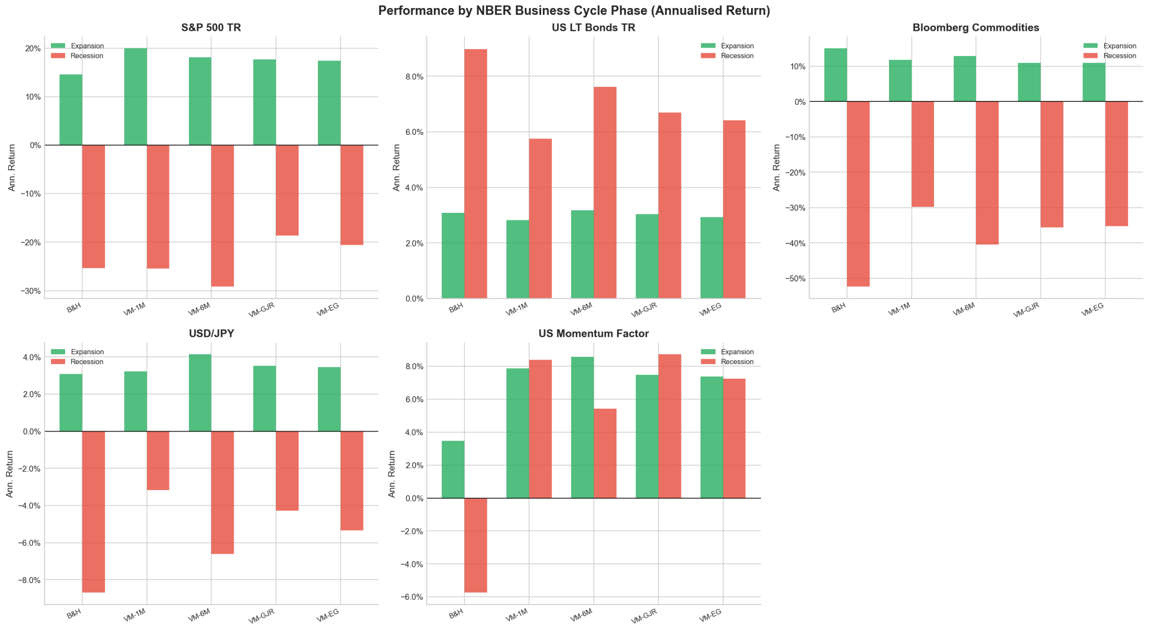

Figure 8: Performance by NBER Business Cycle Phase (Annualised Return). VM strategies provide a meaningful loss reduction during recessions for the S&P 500 (6.8 percentage points), commodities (16.8 percentage points) and the momentum factor (a sign flip from -5.7% to $+8.7\%$). For bonds, the flight-to-quality recession premium is partially sacrificed by the volatility-targeting adjustment.

7.3. Discussion

S&P 500. The recession return improves from -25.4% (Buy and Hold) to -18.6% under GJR-GARCH, a 6.8 percentage point reduction in the annualised loss. Recession volatility falls from 26.5% to 16.0%, confirming that the model successfully de-risks during crises. However, the recession Sharpe ratio is marginally worse (-1.16 against -0.96), a counterintuitive result that arises because the higher expansion volatility of the managed strategy (where leverage is used) raises the denominator. The economically relevant metric, the absolute loss in dollars, is clearly improved.

US LT Bonds. Bonds display the classic flight-to-quality recession premium: Buy and Hold earns 9.0% annualised during recessions against 3.1% during expansions. The managed variants forgo part of this premium (recession return falls to 6.7% under GJR-GARCH) because they de-risk when bond volatility rises. The recession Sharpe under GJR-GARCH (1.13) is slightly higher than Buy and Hold (1.03), thanks to proportional volatility compression, but the difference is well within the sampling-error band of half a Sharpe point.

Bloomberg Commodities. The recession loss shrinks from 52.4% (Buy and Hold) to 35.6% (GJR-GARCH), a 16.8 percentage point reduction. However, the strategies also sacrifice expansion returns (15.0% for Buy and Hold against 10.9% for GJR-GARCH), and the aggregate full-sample result remains unfavourable (Section 6). The volatility-targeting overlay therefore provides genuine recession protection on commodities, but at the cost of giving up expansion returns that more than offset the protection.

USD/JPY. The recession loss is roughly halved, from -8.7% to -4.3% under GJR-GARCH. The expansion return is essentially unchanged. This is one of the cleaner illustrations of the recession-protection mechanism, although the magnitudes are small in absolute terms.

US Momentum Factor. The most striking pattern. The managed strategies turn a negative recession return (-5.7% for Buy and Hold) into a positive one ($+8.7\%$ under GJR-GARCH). The expansion Sharpe also improves, from 0.29 to 0.47. The mechanism is the avoidance of momentum crashes, which are typically triggered by sudden market reversals associated with elevated volatility. Even with the small-sample caveat, the sign flip is too large to be plausibly attributed to sampling noise.

8. Transaction Costs, Turnover and Leverage

8.1. Turnover by Model

Table 20 reports annualised turnover for the four models and the five assets. A turnover of 1.0 means that the entire portfolio weight is replaced once per year on average.

Table 20: Annualised Turnover by Asset and Forecasting Model (transaction cost = 10 bps)

Asset	VM RV 1M	VM RV 6M	VM GJR-GARCH	VM EGARCH
S&P 500 TR	3.73	1.13	3.88	3.68
US LT Bonds TR	3.06	0.68	1.19	1.19
Bloomberg Commodities	3.50	0.78	1.78	1.77
USD/JPY	3.83	0.92	1.66	1.68
US Momentum Factor	2.80	0.80	2.42	2.35

RV 1M generates the highest turnover (between 2.80 and 3.83 across assets), reflecting its sensitivity to month-to-month volatility fluctuations. The GARCH models lie in an intermediate range (between 1.19 and 3.88), while RV 6M consistently achieves the lowest turnover (between 0.68 and 1.13). For the S&P 500, GJR-GARCH turnover (3.88) is marginally higher than that of RV 1M (3.73), a consequence of the granular monthly variance updating implied by the GARCH recursion. The cost penalty at 10 bps for the highest-turnover model is approximately $3.83 \cdot 10 = 38$ bps per year, non-trivial for a low-return asset such as USD/JPY.

8.2. Transaction Cost Sensitivity

Table 21 reports the Sharpe ratio of the GJR-GARCH variant at transaction costs ranging from zero to 100 basis points for every asset. The breakeven cost is defined as the level at which the volatility-managed Sharpe equals the Buy and Hold Sharpe. Table 22 reports the breakeven for each asset.

Table 21: Transaction Cost Sensitivity: VM GJR-GARCH Sharpe Ratio by Cost Level

Asset	B&H	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps	75 bps	100 bps
S&P 500 TR	0.658	0.808	0.796	0.785	0.761	0.737	0.690	0.631	0.572
US LT Bonds TR	0.294	0.275	0.265	0.256	0.237	0.219	0.181	0.134	0.088
Bloomberg Commodities	0.235	0.168	0.165	0.162	0.157	0.151	0.141	0.127	0.113
USD/JPY	0.040	0.132	0.124	0.116	0.100	0.084	0.051	0.010	−0.030
US Momentum Factor	0.066	0.378	0.370	0.363	0.348	0.333	0.303	0.266	0.229

Table 22: Breakeven Transaction Cost by Asset (VM GJR-GARCH vs. Buy and Hold)

Asset	B&H Sharpe	VM Sharpe @ 0 bps	Breakeven TC
S&P 500 TR	0.658	0.808	63.6 bps
US LT Bonds TR	0.294	0.275	$<$ 0 bps (VM never wins)
Bloomberg Commodities	0.235	0.168	$<$ 0 bps (VM never wins)
USD/JPY	0.040	0.132	56.6 bps
US Momentum Factor	0.066	0.378	$>$ 100 bps (VM always wins)

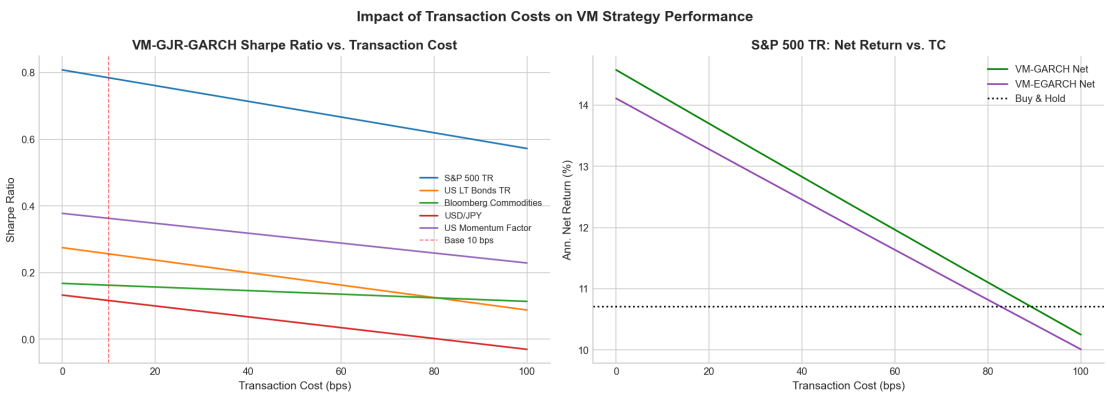

Figure 9: Impact of Transaction Costs on VM Strategy Performance. *Left:* Sharpe ratio against cost for all assets (VM GJR-GARCH); the dashed vertical line marks the 10 bps baseline. *Right:* Net return against cost for the S&P 500 (GJR-GARCH and EGARCH versus the Buy and Hold dotted line). The breakeven cost for equities is approximately 63.6 bps, well above typical institutional execution costs.

Three results merit emphasis. First, for the S&P 500 and USD/JPY, the breakeven cost (between 56 and 64 bps) is well above typical institutional execution costs (5 to 20 bps per Frazzini et al.

(2018)), so the strategy retains a comfortable margin of safety. Second, for US long-term bonds and commodities, the volatility-managed Sharpe is below the Buy and Hold Sharpe even at zero cost (0.275 against 0.294 for bonds; 0.168 against 0.235 for commodities). No cost reduction can rescue the strategy on these assets: the failure is structural, not cost-driven. Third, on the momentum factor the breakeven cost exceeds 100 bps; the volatility-managed variant is preferred at any plausible cost level.

8.3. Turnover Mitigation: Grid Analysis

We evaluate two families of turnover-reduction rules applied to VM GJR-GARCH: threshold bands and exponential moving averages of the raw weights.

Threshold band of width b. The weight is updated only when the new raw weight differs from the previous implemented weight by more than b:

$$w_t = \begin{cases} w_t^{\text{raw}} & \text{if } |w_t^{\text{raw}} - w_{t-1}| > b, \\ w_{t-1} & \text{otherwise.} \end{cases}$$

The new implemented weight is the new raw weight; we do not interpolate to the band edge. We tested four band widths: 5%, 10%, 15% and 20%.

Exponential moving average of span s. The implemented weight is an exponentially weighted moving average of the raw weights with span s, $\tilde{w}_t = \lambda \tilde{w}_{t-1} + (1-\lambda) w_t^{\text{raw}}$ and smoothing factor $\lambda = 2/(s+1) - 1$. We tested four spans: 2, 3, 6 and 12 months.

Table 23 reports the grid: for every asset and rule, the Sharpe ratio, the net annualised return, the turnover, and the deviation from the baseline (no mitigation, raw weight).

Table 23: Turnover Mitigation Grid for VM GJR-GARCH (transaction cost = 10 bps)

Asset	Rule	Sharpe	Net Ret	Ann. TO	Δ Sharpe	Δ TO
	Baseline	0.785	14.14%	3.88	0	0%
	Band 5%	0.783	14.10%	3.84	−0.002	−0.8%
	Band 10%	0.784	14.12%	3.75	−0.000	−3.2%
	Band 15%	0.795	14.28%	3.64	+0.010	−6.0%
S&P 500 TR	Band 20%	0.790	14.14%	3.42	+0.005	−11.9%
	EMA s = 2	0.770	14.10%	2.47	−0.015	−36.1%
	EMA s = 3	0.761	14.14%	1.84	−0.024	−52.6%
	EMA s = 6	0.747	14.25%	1.06	−0.038	−72.6%
	EMA s = 12	0.726	14.19%	0.60	−0.058	−84.5%
	Baseline	0.256	3.18%	1.19	0	0%
	Band 5%	0.266	3.24%	1.11	+0.010	−7.2%
	Band 10%	0.271	3.26%	0.88	+0.015	−26.0%
	Band 15%	0.261	3.20%	0.74	+0.005	−38.4%
US LT Bonds	Band 20%	0.300	3.45%	0.59	+0.044	−50.7%
	EMA s = 2	0.281	3.35%	0.79	+0.025	−33.8%
	EMA s = 3	0.290	3.41%	0.62	+0.034	−47.7%
	EMA s = 6	0.298	3.47%	0.41	+0.042	−65.7%
	EMA s = 12	0.303	3.52%	0.26	+0.047	−78.1%
	Baseline	0.162	1.35%	1.78	0	0%
	Band 5%	0.164	1.40%	1.71	+0.002	−3.9%
	Band 10%	0.160	1.30%	1.56	−0.002	−12.0%
	Band 15%	0.171	1.73%	1.40	+0.009	−21.1%
Bloomberg Comm.	Band 20%	0.172	1.77%	1.26	+0.009	−29.2%
	EMA s = 2	0.148	0.70%	1.17	−0.014	−34.1%
	EMA s = 3	0.152	0.71%	0.89	−0.010	−50.0%
	EMA s = 6	0.165	0.98%	0.54	+0.002	−69.4%
	EMA s = 12	0.164	0.77%	0.32	+0.002	−81.7%
	Baseline	0.116	2.40%	1.66	0	0%
	Band 5%	0.118	2.42%	1.59	+0.002	−4.1%
	Band 10%	0.120	2.44%	1.45	+0.004	−12.6%
	Band 15%	0.129	2.53%	1.23	+0.013	−25.7%
USD/JPY	Band 20%	0.115	2.39%	1.09	−0.001	−34.3%
	EMA s = 2	0.119	2.43%	1.07	+0.003	−35.3%
	EMA s = 3	0.114	2.38%	0.82	−0.002	−50.7%
	EMA s = 6	0.105	2.28%	0.50	−0.011	−69.9%
	EMA s = 12	0.098	2.21%	0.30	−0.019	−82.0%
	Baseline	0.363	6.46%	2.42	0	0%
	Band 5%	0.363	6.47%	2.37	+0.000	−2.4%
	Band 10%	0.351	6.28%	2.29	−0.012	−5.4%
	Band 15%	0.351	6.28%	2.19	−0.012	−9.7%
US Momentum	Band 20%	0.359	6.40%	1.98	−0.004	−18.2%
	EMA s = 2	0.372	6.58%	1.64	+0.009	−32.5%
	EMA s = 3	0.374	6.63%	1.28	+0.012	−47.0%
	EMA s = 6	0.373	6.67%	0.83	+0.010	−65.8%
	EMA s = 12	0.339	6.24%	0.54	−0.023	−77.9%

Three patterns deserve attention. First, for US long-term bonds, every mitigation rule *improves* the Sharpe ratio over the baseline; the largest gain is for Band 20% (Δ Sharpe = $+0.044$ with a 51% turnover reduction). The GJR-GARCH signal is itself noisy on bonds, and dampening it removes more noise than it removes signal. Second, for the S&P 500, the Sharpe ratio is essentially flat across band widths from 5% to 20% (the difference is within ± 0.01), with turnover reductions from 1% to 12%; the EMA rules trade Sharpe for turnover more aggressively, with EMA span 6 cutting turnover by 73% at a cost of -0.038 in Sharpe. Third, for the momentum factor, EMA spans of 2, 3 and 6 deliver a small Sharpe improvement together with turnover reductions of 33% to 66%; this is the rare combination of a Sharpe-positive and TO-negative move.

8.4. Breakeven Transaction Cost under Mitigation

The next question is whether mitigation extends the economic viability of the strategy. Table 24 reports the breakeven cost of VM GJR-GARCH against Buy and Hold under three rules: the baseline (no mitigation), the Band 10% rule, and the EMA span 3 rule.

Table 24: Breakeven Transaction Cost under Mitigation Rules (basis points)

Asset	Baseline	Band 10%	EMA span 3
S&P 500 TR	63.6	65.2	> 100
US LT Bonds TR	< 0	< 0	6.3
Bloomberg Commodities	< 0	< 0	< 0
USD/JPY	56.6	66.1	> 100
US Momentum Factor	> 100	> 100	> 100

On the S&P 500 and USD/JPY, the EMA span 3 rule pushes the breakeven cost above 100 basis points, an improvement of roughly 60% over the baseline. For US long-term bonds, the rule reverses the verdict: the strategy moves from never winning to a breakeven of 6.3 bps, although the practical relevance of this number is limited by the small absolute Sharpe difference. The combination of a moderate EMA smoothing and the GJR-GARCH model therefore extends the cost robustness of the strategy substantially on equities and FX.

8.5. Leverage Sensitivity

Table 25 analyses the sensitivity of VM GJR-GARCH performance to alternative leverage caps. Figure 6 (bottom-right panel) plots the Sharpe ratio against the cap level.

Table 25: Leverage Constraint Sensitivity: VM GJR-GARCH Sharpe Ratios (transaction cost = 10 bps)

Asset	1.0	1.25	1.5	2.0	3.0	No Cap
S&P 500 TR	0.769	0.785	**0.803**	0.785	0.782	0.782
US LT Bonds TR	**0.289**	0.281	0.259	0.256	0.256	0.256
Bloomberg Commodities	**0.226**	0.206	0.174	0.162	0.160	0.160
USD/JPY	0.071	0.091	0.105	**0.116**	0.116	0.116

For equities, the Sharpe ratio is maximised at a cap of 1.5 (Sharpe 0.803), marginally above the 2.0 baseline. The intuition is the trade-off between capturing low-volatility leverage opportunities and incurring additional rebalancing costs at the boundary; the optimum lies in between. Constraining leverage to 1.0 eliminates leverage-related cost drag but sacrifices return, reducing the equity Sharpe to 0.769. For bonds and commodities, tighter constraints improve risk-adjusted performance monotonically: the additional leverage employed by GJR-GARCH on these assets does not generate sufficient return to compensate for its costs. For USD/JPY, the relation is inverse: the unmanaged weight at 1.0 has too little exposure to capture the modest signal, and the Sharpe rises monotonically with the cap up to 2.0.

In practical terms, a 1.5 cap is optimal for equities, 1.0 for bonds and commodities, and 2.0 for USD/JPY. These choices preserve essentially all of the strategy's risk-adjusted performance while limiting operational and margin risk.

9. Summary and Recommendations

9.1. Synthesis of the Evidence

This study evaluates volatility-targeting strategies on five asset classes over twenty-six years, using four forecasting models and a look-ahead-bias-free implementation. The evidence supports a more cautious conclusion than the early literature would suggest.

The four conditions identified in Section 1 are useful to organise the results:

1. *Forecast quality.* The S&P 500, the momentum factor and US LT bonds have a Mincer-Zarnowitz R^2 above 0.40 for at least one model; USD/JPY does not. Forecast quality alone, however, does not guarantee economic value: commodities have an R^2 above 0.40 for GJR-GARCH and EGARCH, yet the strategy destroys value on commodities.
2. *Return-volatility dynamics.* The strategy benefits assets where the conditional Sharpe ratio is decreasing in volatility. This holds for the momentum factor in our sample, where the largest losses occur during the highest volatility months. It fails for US long-term bonds, where the flight-to-quality effect reverses the relationship.

3. *Transaction costs.* For equities and FX, the breakeven cost is well above institutional execution costs and the strategy retains a margin of safety. For bonds and commodities, the strategy is below Buy and Hold even at zero cost.

4. *Statistical significance.* This is where the evidence is the most restrictive. Only on the momentum factor do the Sharpe ratio differences pass the Jobson-Korkie and bootstrap tests at the 5% level; the equity improvement, while economically meaningful, is not statistically distinguishable from zero in our sample.

A central methodological lesson is that the look-ahead bias in the target volatility is quantitatively smaller in our sample than Liu et al. (2019) document for the US equity factor universe (an average inflation of -2.4% across all twenty cells). The statistical-significance issue, however, survives in our data: many improvements that look large in point-estimate terms do not survive a formal test. This is the more durable of the two critiques of the volatility-management literature.

9.2. Recommendations to the Volatility Management Board

Recommendation 1: Adopt for the momentum factor. The improvement is the only one in this study that is statistically significant at the 5% level under both Jobson-Korkie-Memmel and bootstrap tests, on all four forecasting models (the bootstrap p-values range from 0.003 to 0.015). The point estimate is also the largest: Δ Sharpe = $+0.297$ for GJR-GARCH against Buy and Hold, with the worst drawdown reduced from 61% to 38% and the recovery time shortened from a never-recovered position to 64 months. The Fama-French alphas of 7.5% to 8.7% per year ($p < 0.05$) confirm that this is genuine timing skill rather than a static factor tilt. *Suggested specification:* VM GJR-GARCH or VM RV 6M (the latter has the lowest turnover at 0.80 and the highest Sharpe at 0.400), with EMA smoothing of span 3 (which preserves the Sharpe) and a leverage cap of 2.0.

Recommendation 2: Adopt for the S&P 500 with caveats. The point estimate is favourable (Δ Sharpe = $+0.126$ for GJR-GARCH, with the maximum drawdown reduced from 51% to 37% and the recovery time from 37 to 20 months), the breakeven cost is comfortable (about 64 bps), and the strategy's recession-loss reduction is meaningful (6.8 percentage points). However, the Sharpe ratio difference is not statistically significant ($p = 0.30$ bootstrap, 95% confidence interval $[-0.10, +0.35]$), and the Fama-French alpha is not statistically different from zero ($p = 0.13$ on FF3). The Board should be aware that the statistical evidence is suggestive rather than conclusive, and that out-of-sample performance is therefore more uncertain than the point estimates would imply. *Suggested specification:* VM GJR-GARCH with EMA smoothing of span 3 (pushes the breakeven cost above 100 bps) and a leverage cap of 1.5.

Recommendation 3: Do not adopt for commodities. The Sharpe ratio under volatility management is below the Buy and Hold Sharpe at every positive cost level, and the breakeven cost is below zero. The structural combination of high base volatility, weak forecast quality and the secular commodity-price decline of the post-2008 period leaves the strategy without an edge. We recommend that the Board manage commodity risk through position sizing and cross-asset diversification rather than through volatility targeting on the commodity sleeve.

Recommendation 4: Do not adopt for US long-term bonds. The flight-to-quality dynamic means that bond volatility spikes coincide with positive return episodes, so the strategy de-risks at the moments when bonds are most useful. The Sharpe difference is modest (-0.038 for GJR-GARCH) but consistent across all four models, and the breakeven cost is below zero. A marginal improvement in recession Sharpe (1.13 against 1.03 for Buy and Hold) is within the sampling-error band and should not be relied upon. The Band 20% and EMA span 12 mitigation rules *do* improve the bond Sharpe somewhat by removing noise, but the gains remain too small to constitute a compelling case for adoption.

Recommendation 5: Marginal adoption for USD/JPY. The improvement is positive but modest (Δ Sharpe = $+0.076$ for GJR-GARCH, $+0.101$ for RV 6M). The Sharpe difference is not statistically significant under either test (p-values between 0.14 and 0.29). Given the weak FX forecastability, the strategy adds value primarily through drawdown reduction (from 37.1% to 35.4%). *Suggested specification:* VM RV 6M (highest Sharpe at 0.142, lowest turnover at 0.92), or a combination of GJR-GARCH with Band 15% mitigation.

Recommendation 6: Model selection. Choose GJR-GARCH as the primary model on three grounds: superior RMSE and R^2 for equities and momentum, an economically interpretable leverage-effect parameter $\gamma > 0$, and lower turnover than RV 1M. EGARCH serves as a robustness check and is preferred on pure forecast-accuracy grounds for bonds, commodities and USD/JPY; the net Sharpe difference between the two GARCH variants is statistically insignificant on every asset except US LT bonds, where the difference is too small to be of practical relevance.

Recommendation 7: Implementation overlays. For every adopted strategy, an EMA smoothing of span 3 reduces turnover by 30% to 50% with minimal Sharpe sacrifice. On equities and FX, this is also sufficient to push the breakeven transaction cost above 100 basis points. For bonds (if eventually adopted), a Band 20% rule delivers the largest Sharpe improvement. We recommend that mitigation rules be standardised across the adopted strategies to simplify implementation and monitoring.

Recommendation 8: Multi-asset implementation. The equal-weight portfolio of the five GJR-GARCH variants achieves a Sharpe ratio (0.673) higher than any single-asset variant in our sample, with a maximum drawdown of 19.5%. Volatility timing and cross-asset diversification are complements, not substitutes. Even if the Board chooses not to adopt every single-asset strategy in isolation (commodities, bonds), retaining them in a diversified multi-asset portfolio may still be beneficial, because their volatility-timed returns are nearly orthogonal to the equity and momentum sleeves. A formal portfolio optimisation is left for future work.

9.3. Limitations and Future Research

Several limitations qualify our results. First, the 2000–2025 sample is dominated by two major crises (2008 and 2020) and includes the extended low-interest-rate environment of 2010–2021; the strategy's value-add over a different sample window may differ from the full-sample estimates (Liu et al., 2019). Second, daily rebalancing within each month is assumed, which may not match the operational constraints of real-world implementation. Third, the transaction cost model is simple: linear, one-way, with no market impact or bid-ask spread non-linearities for large positions. Fourth, the multi-asset analysis is restricted to equal weighting; a formal multi-asset optimisation with time-varying covariance estimation is left for future work. Finally, our Fama-French risk model applies only to the S&P 500 and the momentum factor; bond, commodity and FX assets would require their own multi-factor specifications (term and default factors, commodity carry and momentum, FX carry and value), which we do not implement.

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